HVAC DESIGN CALCULATION REPORT

DDA MIG Housing Scheme 182A (test run under KSA codes)

Riyadh, Saudi Arabia

Building type: apartment - 4 floor(s)

Code basis: Saudi Building Code + ASHRAE | Units: SI

Issue: P0 (preliminary) 2026-10-15

PRELIMINARY DESIGN - FOR COORDINATION ONLY. To be checked and stamped by a qualified HVAC engineer before construction.

1. Design basis

1.1 Codes and standards

Ref	Code / standard
SBC501	SBC 501 Saudi Mechanical Code (2018, as amended)
SBC601	SBC 601 Saudi Energy Conservation Code - Commercial
SBC602	SBC 602 Saudi Energy Conservation Code - Residential
A62.1	ANSI/ASHRAE Standard 62.1-2019 Ventilation for Acceptable IAQ
A90.1	ANSI/ASHRAE/IES Standard 90.1-2019 Energy Standard
AF	ASHRAE Handbook - Fundamentals (CLTD/SCL/CLF method 1997 Ch.28; climate data 2021 Ch.14)
SMACNA	SMACNA HVAC Duct Construction Standards - Metal and Flexible, 3rd ed. 2005

1.2 Design conditions

Parameter	Value	Basis
Outdoor summer design (0.4 %)	45.6 °C DB / 19.4 °C MCWB, W = 4.4 g/kg	ASHRAE 2021 Ch.14 (0.4% cool / 99.6% heat) - OERK
Daily range	14.6 K	ASHRAE 2021 Ch.14 (0.4% cool / 99.6% heat) - OERK
Outdoor winter design (99.6 %)	6.6 °C	ASHRAE 2021 Ch.14 (0.4% cool / 99.6% heat) - OERK
Indoor cooling	24.0 °C, 50 % RH (W = 10.0 g/kg)	SBC 602 §302 / SBC 601 §302.1 (design conditions); ASHRAE Fundamentals 2021 Ch.14 (0.4 % cooling, 99.6 % heating)
Indoor heating	21.0 °C	SBC 602 §302 / SBC 601 §302.1 (design conditions); ASHRAE Fundamentals 2021 Ch.14 (0.4 % cooling, 99.6 % heating)
Site elevation / pressure	612 m / 94.19 kPa; air factors 1.143 W/(L/s·K), 2798 W/(L/s·kg/kg)	ASHRAE Fundamentals Ch.1
Latitude / design month	24.71° / July (CLTD tables)	ASHRAE Fundamentals 1997 Ch.28 CLTD/SCL/CLF method, Tables 30-36

Code reference: SBC 602 §302 / SBC 601 §302.1 (design conditions); ASHRAE Fundamentals 2021 Ch.14 (0.4 % cooling, 99.6 % heating)

1.3 Building envelope

Element	U-value / property	Basis
External wall	0.40 W/m²K (CLTD wall group D)	SBC602
Roof	0.20 W/m²K (roof_insulated_concrete)	SBC602
Glazing	2.67 W/m²K, SHGC 0.25 (SC 0.29)	SBC602
Slab edge (heating)	F = 1.2 W/m·K	SBC602

Code reference: SBC 602 Ch.4 prescriptive envelope table / SBC 601 Ch.5 (max U-factor, SHGC)

1.4 Internal loads

Space type	Occupancy	Activity	Lighting W/m²	Equipment W/m² + fixed	OA category	Exhaust	Cooled
bath	0 persons	seated	8	0 + 0 W		toilet_private	no
bedroom	2 persons	sleeping	8	5 + 0 W	dwelling		yes
dining	2.5 m²/person	seated	10	3 + 0 W	dwelling		yes
kitchen	2 persons	light_work	10	60 + 800 W	dwelling	kitchen_residential	yes
outdoor	0 persons	seated	0	0 + 0 W			no

shaft	0 persons	seated	0	0 + 0 W			no

Code reference: ASHRAE Fundamentals 2021 Ch.18 Table 1 (people), Tables 2-3 (lighting), Tables 5-9 (equipment)

1.5 Assumptions

All assumptions are logged in assumptions.md. Items marked 'confirm' must be confirmed by the engineer of record.

ID	Step	Assumption	Source	Status
A001	A-read	Window W1 size 600x600 mm given by user (not in schedule)	user	default
A002	A-read	Page 1 scale 1:139.0 (user, confidence high)	scale_detect.py	default
A003	Gate1	Page 1 TERRACE: measured 6.06 m² vs plan text 7.36 m² (>12% difference). Which is right? -> USER ANSWER: use measured	user	default
A004	Gate1	Page 1 DINNING /LOUNGE: measured 22.82 m² vs plan text 11.14 m² (>12% difference). Which is right? -> USER ANSWER: drawing room + dining/lounge are open plan - one living zone ~18 m2 (measured)	user	default
A005	Gate1	Page 1 KITCHEN: measured 4.24 m² vs plan text 4.85 m² (>12% difference). Which is right? -> USER ANSWER: use measured	user	default
A006	Gate1	Page 1 BATH: measured 1.59 m² vs plan text 1.85 m² (>12% difference). Which is right? -> USER ANSWER: use measured	user	default
A007	Gate1	Page 1 SHAFT: measured 2.69 m² vs plan text 3.11 m² (>12% difference). Which is right? -> USER ANSWER: use measured	user	default
A008	Gate1	Page 1 ROOM: measured 7.57 m² vs plan text 9.06 m² (>12% difference). Which is right? -> USER ANSWER: use measured (traced drawing)	user	default
A009	A-read	Page 1: 36 labelled cells ignored as non-rooms (schedule/title/grid text): ['3.TERRACE', 'CB', 'D D A', 'DILSHAD GARDEN', 'DINNING /LOUNGE', 'Digitized By', 'EAST ZONE, VIKAS MINAR, D.D.A.', 'FG1', 'GROUND AND FIRST', 'JAHANGIRPURI', 'PASHCHIM PURI', 'PITAMPURA (D&P)', 'SHALIMAR BHAGH (P)', 'STANDARD DESIGN FOR MIG', 'Sch.no. 182A']	pdf_reader.py	default
A010	Gate1	Page 1: 15 enclosed space(s) without a name (areas m²: [1.61, 1.71, 1.71, 3.26, 2.23, 1.71, 1.7, 1.7, 1.7, 2.26, 3.29, 1.71, 1.71, 2.66, 1.61]). Name them or confirm they are voids. -> USER ANSWER: small unnamed spaces are flat entrance lobbies/passages - not cooled, transfer air	user	default
A011	A-read	Page 1: 'GROUND FLOOR PLAN' plan separated from the other plan(s) on the sheet by nearest title	pdf_reader.py	default

A012	A-read	Page 1: 'FIRST FLOOR PLAN' plan separated from the other plan(s) on the sheet by nearest title	pdf_reader.py	default
A013	A-read	Page 2 scale 1:139.0 (user, confidence high)	scale_detect.py	default
A014	Gate1	Page 2 ROOM: measured 10.18 m² vs plan text 9.06 m² (>12% difference). Which is right? -> USER ANSWER: use measured	user	default
A015	Gate1	Page 2 DINNING /LOUNGE: measured 22.92 m² vs plan text 11.14 m² (>12% difference). Which is right? -> USER ANSWER: same - one living zone	user	default
A016	Gate1	Page 2 KITCHEN: measured 5.74 m² vs plan text 4.85 m² (>12% difference). Which is right? -> USER ANSWER: use measured	user	default
A017	Gate1	Page 2: one enclosed space carries several names ['SHAFT', 'SHAFT'] - an opening may be unsealed. Split or confirm. -> USER ANSWER: two shaft labels in one light well - one shaft	user	default
A018	Gate1	Page 2 TERRACE: measured 9.24 m² vs plan text 7.36 m² (>12% difference). Which is right? -> USER ANSWER: use measured	user	default
A019	Gate1	Page 2 SHAFT: measured 3.65 m² vs plan text 3.11 m² (>12% difference). Which is right? -> USER ANSWER: use measured	user	default
A020	Gate1	Page 2 BATH: measured 2.08 m² vs plan text 1.85 m² (>12% difference). Which is right? -> USER ANSWER: use measured	user	default
A021	A-read	Page 2: 33 labelled cells ignored as non-rooms (schedule/title/grid text): ['CB', 'D D A', 'DILSHAD GARDEN', 'Digitized By', 'EAST ZONE, VIKAS MINAR, D.D.A.', 'FG1', 'JAHANGIRPURI', 'PASHCHIM PURI', 'PITAMPURA (D&P)', 'SECOND AND THIRD', 'SHALIMAR BHAGH (P)', 'STANDARD DESIGN FOR MIG', 'Sch.no. 182A', 'TERRACE', 'WING']	pdf_reader.py	default
A022	Gate1	Page 2: 9 enclosed space(s) without a name (areas m²: [2.41, 2.42, 3.46, 3.53, 3.7, 3.44, 3.52, 2.42, 2.42]). Name them or confirm they are voids. -> USER ANSWER: same - passages	user	default
A023	A-read	Page 2: 'SECOND FLOOR PLAN' plan separated from the other plan(s) on the sheet by nearest title	pdf_reader.py	default
A024	A-read	Page 2: 'THIRD FLOOR PLAN' plan separated from the other plan(s) on the sheet by nearest title	pdf_reader.py	default
A025	A-read	Page 3 scale 1:139.0 (user, confidence high)	scale_detect.py	default
A026	Gate1	Page 3 SHAFT: measured 3.65 m² vs plan text 3.11 m² (>12% difference). Which is right? -> USER ANSWER: use measured	user	default
A027	A-read	Page 3: 22 labelled cells ignored as non-rooms (schedule/title/grid text): ['D D A',	pdf_reader.py	default

		'Digitized By', 'SLOPE', 'STANDARD DESIGN FOR MIG', 'Sch.no. 182A', 'WING', 'WITH SA VIDE LETTER NO.ACE/D/III/S.D/9/E', 'add. ch. arch.', 'arch. asstt.', 'asstt. arch.', 'asstt. dir.', 'chief arch.', 'deputy director']		
A028	Gate1	Page 3: 8 enclosed space(s) without a name (areas m²: [2.4, 2.32, 3.5, 2.91, 2.29, 2.13, 3.53, 2.42]). Name them or confirm they are voids. -> USER ANSWER: same - passages / mumty	user	default
A029	Gate1	Second Floor: 4 rooms copied from Third Floor where the plan did not read (typical flats): SF-34 BALCONY (TF-02), SF-35 BALCONY (TF-03), SF-36 BALCONY (TF-32), SF-37 BALCONY (TF-33)	user instruction	CONFIRM
A030	B-loads	Outdoor design 45.6 °C DB / 19.4 °C MCWB (0.4 %), daily range 14.6 K, heating 6.6 °C (99.6 %), elevation 612.0 m (ASHRAE 2021 Ch.14 (0.4% cool / 99.6% heat) - OERK, approximate - verify)	references/climate_data.csv	default
A031	B-loads	Indoor design 24.0 °C / 50 % RH cooling, 21.0 °C heating	SBC 602 §302 / SBC 601 §302.1 (design conditions); ASHRAE Fundamentals 2021 Ch.14 (0.4 % cooling, 99.6 % heating)	default
A032	B-loads	Envelope U-values = code maximum (SBC602: wall 0.4, roof 0.2, glazing U 2.67 / SHGC 0.25 W/m²K) - actual build-up not given	SBC 602 Ch.4 prescriptive envelope table / SBC 601 Ch.5 (max U-factor, SHGC)	CONFIRM
A033	B-loads	Cooling loads by CLTD/SCL/CLF; CLTD tables for wall group D, roof 'roof_insulated_concrete', latitude row nearest 24.71°, July; colour factor K=0.83 (medium walls), roof K=1.0; glass with interior shading=True	ASHRAE Fundamentals 1997 Ch.28 CLTD/SCL/CLF method, Tables 30-36	CONFIRM
A034	B-loads	CLTD/LM/SHGF/CLF tables in references/cltd_tables.json are transcribed-approximate - verify against licensed ASHRAE 1997 Ch.28 tables	references/cltd_tables.json	CONFIRM
A035	B-loads	CLF for people/lights/equipment = 1.0 (24 h operation, no storage credit)	ASHRAE Fundamentals 1997 Ch.28 CLTD/SCL/CLF method, Tables 30-36	default
A036	B-loads	Partitions/floors between conditioned and exhaust-only rooms (baths, stores) not counted: those rooms are near room temperature	designer judgement	default
A037	B-loads	Outdoor humidity ratio 4.4 g/kg < indoor 10.0 g/kg at design: outdoor-air latent load taken as 0 (conservative); check a dehumidification design day	ASHRAE Fundamentals 2021 Ch.14 (DP design)	CONFIRM
A038	B-loads	Conditioned/unconditioned overrides from the user: by type {'kitchen': False}, by room {}	user instruction	default
A039	B-loads	Supply air 13.0 °C (room-supply ΔT 11.0 K) for airflow	designer default	default
A040	B-loads	Infiltration by air-change method: {'one_exterior_side': 0.5, 'two_plus_exterior_sides': 0.75, 'no_exterior': 0.0} ACH (residential)	ASHRAE Fundamentals 2021 Ch.16 §'Residential air change method'	default
A041	B-loads	Occupancy, lighting and equipment densities from references/space_types.json	ASHRAE Fundamentals 2021 Ch.18 Table 1 (people), Tables 2-3 (lighting), Tables 5-9 (equipment)	default

A042	B-zoning	Zoning: own zone for ['majlis', 'master_bedroom']; other ducted rooms grouped by position per floor up to 17.6 kW per unit (incl. safety factor)	references/design_defaults.json zoning	CONFIRM
A043	B-zoning	Rooms exhaust-only (not cooled; make-up by transfer air): GF-04, GF-05, GF-06, GF-08, GF-09, GF-17, GF-18, GF-19, GF-21, GF-22, GF-23, GF-33, GF-34, GF-36, GF-37, FF-07, FF-08, FF-09, FF-11, FF-12, FF-13, FF-18, FF-19, FF-20, FF-22, FF-23, FF-24, SF-05, SF-07, SF-09, SF-11, SF-12, SF-13, SF-14, SF-20, SF-21, SF-23, SF-25, SF-27, SF-29, TF-07, TF-09, TF-11, TF-13, TF-14, TF-15, TF-19, TF-20, TF-22, TF-24, TF-26, TF-28	references/space_types.json	CONFIRM
A044	B-vent	Zone air distribution effectiveness Ez = 0.8 (heat-pump heating with ceiling supply of warm air)	ASHRAE 62.1-2019 §6.2.2.2 Table 6-2 (Ez)	default
A045	B-vent	Exhaust fans sized for the intermittent (rated) rate; continuous operation would allow the lower rate	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)	default
A046	B-vent	Building OA 891 L/s < exhaust 1725 L/s (intermittent): building slightly negative when all fans run; acceptable for intermittent fans, else raise OA	designer judgement	CONFIRM
A047	B-equip	Safety factor 10% on cooling and 10% on heating, applied to coil loads at selection (shown in tables)	Designer allowance - not a code requirement	CONFIRM
A048	B-equip	Nominal (T1) capacities derated x0.88 for 45.6 °C outdoor design (T3) - replace with brand T3 data	references/equipment_catalog.json t3_derate	CONFIRM
A049	B-equip	Equipment from GENERIC catalogue (references/equipment_catalog.json) - replace with the approved brand	references/equipment_catalog.json	CONFIRM
A050	B-equip	Minimum supply per served room 30 L/s (smallest diffuser)	references/air_terminals.json	default
A051	B-equip	IU-GF-01 (WS-09) is 194 % above requirement - smallest catalogue size; consider combining zones or a smaller (VRF/wall) unit	designer note	CONFIRM
A052	B-equip	IU-GF-02 (WS-09) is 215 % above requirement - smallest catalogue size; consider combining zones or a smaller (VRF/wall) unit	designer note	CONFIRM
A053	B-equip	IU-GF-03 (WS-09) is 193 % above requirement - smallest catalogue size; consider combining zones or a smaller (VRF/wall) unit	designer note	CONFIRM
A054	B-equip	IU-GF-04 (WS-09) is 184 % above requirement - smallest catalogue size; consider combining zones or a smaller (VRF/wall) unit	designer note	CONFIRM
A055	B-equip	IU-GF-05 (WS-09) is 192 % above requirement - smallest catalogue size; consider combining zones or a smaller (VRF/wall) unit	designer note	CONFIRM
A056	B-equip	IU-GF-06 (WS-09) is 226 % above requirement - smallest catalogue size; consider combining zones or a smaller (VRF/wall) unit	designer note	CONFIRM
A057	B-equip	IU-GF-07 (WS-09) is 204 % above requirement - smallest catalogue size;	designer note	CONFIRM

		consider combining zones or a smaller (VRF/wall) unit		
A058	B-equip	IU-GF-08 (WS-09) is 178 % above requirement - smallest catalogue size; consider combining zones or a smaller (VRF/wall) unit	designer note	CONFIRM
A059	B-equip	IU-GF-09 (DS-18) is 76 % above requirement - smallest catalogue size; consider combining zones or a smaller (VRF/wall) unit	designer note	CONFIRM
A060	B-equip	Z-GF-09: no corridor/service room next to the zone - ducted unit in the DINNING /LOUNGE (GF-25) ceiling void, access panel required	designer judgement	CONFIRM
A061	B-equip	IU-GF-10 (DS-18) is 72 % above requirement - smallest catalogue size; consider combining zones or a smaller (VRF/wall) unit	designer note	CONFIRM
A062	B-equip	Z-GF-10: no corridor/service room next to the zone - ducted unit in the DINNING /LOUNGE (GF-26) ceiling void, access panel required	designer judgement	CONFIRM
A063	B-equip	IU-GF-11 (DS-18) is 81 % above requirement - smallest catalogue size; consider combining zones or a smaller (VRF/wall) unit	designer note	CONFIRM
A064	B-equip	Z-GF-11: no corridor/service room next to the zone - ducted unit in the DINNING /LOUNGE (GF-31) ceiling void, access panel required	designer judgement	CONFIRM
A065	B-equip	IU-GF-12 (DS-18) is 78 % above requirement - smallest catalogue size; consider combining zones or a smaller (VRF/wall) unit	designer note	CONFIRM
A066	B-equip	Z-GF-12: no corridor/service room next to the zone - ducted unit in the DINNING /LOUNGE (GF-32) ceiling void, access panel required	designer judgement	CONFIRM
A067	B-equip	IU-FF-01 (WS-09) is 95 % above requirement - smallest catalogue size; consider combining zones or a smaller (VRF/wall) unit	designer note	CONFIRM
A068	B-equip	IU-FF-02 (WS-09) is 129 % above requirement - smallest catalogue size; consider combining zones or a smaller (VRF/wall) unit	designer note	CONFIRM
A069	B-equip	IU-FF-03 (WS-09) is 178 % above requirement - smallest catalogue size; consider combining zones or a smaller (VRF/wall) unit	designer note	CONFIRM
A070	B-equip	IU-FF-04 (WS-09) is 192 % above requirement - smallest catalogue size; consider combining zones or a smaller (VRF/wall) unit	designer note	CONFIRM
A071	B-equip	IU-FF-05 (WS-09) is 177 % above requirement - smallest catalogue size; consider combining zones or a smaller (VRF/wall) unit	designer note	CONFIRM
A072	B-equip	IU-FF-06 (WS-09) is 191 % above requirement - smallest catalogue size; consider combining zones or a smaller (VRF/wall) unit	designer note	CONFIRM
A073	B-equip	IU-FF-07 (WS-09) is 97 % above	designer note	CONFIRM

		requirement - smallest catalogue size; consider combining zones or a smaller (VRF/wall) unit		
A074	B-equip	IU-FF-08 (WS-09) is 131 % above requirement - smallest catalogue size; consider combining zones or a smaller (VRF/wall) unit	designer note	CONFIRM
A075	B-equip	IU-FF-09 (DS-18) is 81 % above requirement - smallest catalogue size; consider combining zones or a smaller (VRF/wall) unit	designer note	CONFIRM
A076	B-equip	Z-FF-09: no corridor/service room next to the zone - ducted unit in the DINNING /LOUNGE (FF-04) ceiling void, access panel required	designer judgement	CONFIRM
A077	B-equip	IU-FF-10 (DS-18) is 78 % above requirement - smallest catalogue size; consider combining zones or a smaller (VRF/wall) unit	designer note	CONFIRM
A078	B-equip	Z-FF-10: no corridor/service room next to the zone - ducted unit in the DINNING /LOUNGE (FF-05) ceiling void, access panel required	designer judgement	CONFIRM
A079	B-equip	IU-FF-11 (DS-18) is 76 % above requirement - smallest catalogue size; consider combining zones or a smaller (VRF/wall) unit	designer note	CONFIRM
A080	B-equip	Z-FF-11: no corridor/service room next to the zone - ducted unit in the DINNING /LOUNGE (FF-26) ceiling void, access panel required	designer judgement	CONFIRM
A081	B-equip	IU-FF-12 (DS-18) is 76 % above requirement - smallest catalogue size; consider combining zones or a smaller (VRF/wall) unit	designer note	CONFIRM
A082	B-equip	Z-FF-12: no corridor/service room next to the zone - ducted unit in the DINNING /LOUNGE (FF-27) ceiling void, access panel required	designer judgement	CONFIRM
A083	B-equip	IU-SF-01 (WS-09) is 150 % above requirement - smallest catalogue size; consider combining zones or a smaller (VRF/wall) unit	designer note	CONFIRM
A084	B-equip	IU-SF-02 (WS-09) is 157 % above requirement - smallest catalogue size; consider combining zones or a smaller (VRF/wall) unit	designer note	CONFIRM
A085	B-equip	IU-SF-03 (WS-09) is 217 % above requirement - smallest catalogue size; consider combining zones or a smaller (VRF/wall) unit	designer note	CONFIRM
A086	B-equip	IU-SF-04 (WS-09) is 130 % above requirement - smallest catalogue size; consider combining zones or a smaller (VRF/wall) unit	designer note	CONFIRM
A087	B-equip	IU-SF-05 (WS-09) is 153 % above requirement - smallest catalogue size; consider combining zones or a smaller (VRF/wall) unit	designer note	CONFIRM
A088	B-equip	IU-SF-06 (WS-09) is 130 % above requirement - smallest catalogue size; consider combining zones or a smaller (VRF/wall) unit	designer note	CONFIRM

A089	B-equip	IU-SF-07 (WS-09) is 152 % above requirement - smallest catalogue size; consider combining zones or a smaller (VRF/wall) unit	designer note	CONFIRM
A090	B-equip	IU-SF-08 (WS-09) is 132 % above requirement - smallest catalogue size; consider combining zones or a smaller (VRF/wall) unit	designer note	CONFIRM
A091	B-equip	IU-SF-09 (WS-09) is 138 % above requirement - smallest catalogue size; consider combining zones or a smaller (VRF/wall) unit	designer note	CONFIRM
A092	B-equip	Z-SF-10: no corridor/service room next to the zone - ducted unit in the DINNING /LOUNGE (SF-06) ceiling void, access panel required	designer judgement	CONFIRM
A093	B-equip	Z-SF-11: no corridor/service room next to the zone - ducted unit in the DINNING /LOUNGE (SF-10) ceiling void, access panel required	designer judgement	CONFIRM
A094	B-equip	Z-SF-12: no corridor/service room next to the zone - ducted unit in the DINNING /LOUNGE (SF-24) ceiling void, access panel required	designer judgement	CONFIRM
A095	B-equip	Z-SF-13: no corridor/service room next to the zone - ducted unit in the DINNING /LOUNGE (SF-28) ceiling void, access panel required	designer judgement	CONFIRM
A096	B-equip	IU-TF-01 (WS-09) is 98 % above requirement - smallest catalogue size; consider combining zones or a smaller (VRF/wall) unit	designer note	CONFIRM
A097	B-equip	IU-TF-02 (WS-09) is 118 % above requirement - smallest catalogue size; consider combining zones or a smaller (VRF/wall) unit	designer note	CONFIRM
A098	B-equip	IU-TF-03 (WS-09) is 116 % above requirement - smallest catalogue size; consider combining zones or a smaller (VRF/wall) unit	designer note	CONFIRM
A099	B-equip	IU-TF-04 (WS-09) is 137 % above requirement - smallest catalogue size; consider combining zones or a smaller (VRF/wall) unit	designer note	CONFIRM
A100	B-equip	IU-TF-05 (WS-09) is 115 % above requirement - smallest catalogue size; consider combining zones or a smaller (VRF/wall) unit	designer note	CONFIRM
A101	B-equip	IU-TF-06 (WS-09) is 135 % above requirement - smallest catalogue size; consider combining zones or a smaller (VRF/wall) unit	designer note	CONFIRM
A102	B-equip	IU-TF-07 (WS-09) is 99 % above requirement - smallest catalogue size; consider combining zones or a smaller (VRF/wall) unit	designer note	CONFIRM
A103	B-equip	IU-TF-08 (WS-09) is 103 % above requirement - smallest catalogue size; consider combining zones or a smaller (VRF/wall) unit	designer note	CONFIRM
A104	B-equip	Z-TF-09: no corridor/service room next to the zone - ducted unit in the DINNING /LOUNGE (TF-08) ceiling void, access	designer judgement	CONFIRM

		panel required		
A105	B-equip	Z-TF-10: no corridor/service room next to the zone - ducted unit in the DINNING /LOUNGE (TF-12) ceiling void, access panel required	designer judgement	CONFIRM
A106	B-equip	Z-TF-11: no corridor/service room next to the zone - ducted unit in the DINNING /LOUNGE (TF-23) ceiling void, access panel required	designer judgement	CONFIRM
A107	B-equip	Z-TF-12: no corridor/service room next to the zone - ducted unit in the DINNING /LOUNGE (TF-27) ceiling void, access panel required	designer judgement	CONFIRM
A108	B-equip	Wall-split outdoor units on brackets on the facade/courtyard wall just outside the served room (architect to approve appearance); ducted-unit ODUs on the roof	user preference	CONFIRM
A109	B-equip	Outdoor units grouped on the roof near the riser/shaft with 0.6 m service clearance; check structure, parapet height and acoustic screening	designer judgement	CONFIRM
A110	B-ducts	Ducts sized by equal friction 0.8 Pa/m with velocity limits {'main': 5.0, 'branch': 3.5, 'runout': 2.5, 'return': 3.5, 'fresh_air': 4.0, 'exhaust': 4.0} m/s; max duct depth 250 mm from ceiling void	ASHRAE Fundamentals 2021 Ch.21 (equal friction method, Eq.18-19 Altshul-Tsal)	default
A111	B-ducts	Fittings allowed as +50% equivalent length; unit plenum/filter 40.0 Pa	designer default	CONFIRM
A112	B-ducts	Duct routes are schematic trunk-and-branch lines over the plan; coordinate with structure (beams), lighting and other services	designer judgement	CONFIRM
A113	B-ducts	Return air from rooms via door undercut / door transfer grilles to the return grille near the unit	ASHRAE Fundamentals 2021 Ch.21 (equal friction method, Eq.18-19 Altshul-Tsal)	CONFIRM
A114	B-pipes	Refrigerant pipe sizes from GENERIC table (references/pipe_sizing.json) - replace with the selected brand's table	SBC 501 Ch.11 §1107 (refrigerant piping); ASHRAE 15; manufacturer piping tables	CONFIRM
A115	B-pipes	Refrigerant length = route x (1+10%) + vertical rise to roof (+1.5 m allowance)	designer default	default
A116	B-pipes	Condensate to nearest wet-room floor drain / stack with trap; min 25 mm, slope 1.0 %	SBC 501 §307.2.1 (disposal), §307.2.2 Table 307.2.2 (drain size), slope >= 1 %	CONFIRM

2. Room schedule (Gate 1 approved)

ID	Room	Type	Area m²	Ceiling m	Exterior walls (orient. m)	Windows	Roof exp.	Cooled
GF-01	VER.	outdoor	3.58	2.75	N 3.05, E 1.17	W3A E 0.575x1.175	100%	no
GF-02	ROOM	bedroom	8.12	2.75	N 3.02	W3A N 0.575x1.175	40%	yes
GF-03	ROOM	bedroom	8.13	2.75	N 2.98	-	41%	yes
GF-04	KITCHEN	kitchen	4.30	2.75	S 1.63	-	0%	yes
GF-05	BATH	bath	1.59	2.75	E 1.19	-	0%	no
GF-06	WC	bath	1.57	2.75	N 0.92	W1 N 1.2x1.54	0%	no
GF-07	SHAFT	shaft	2.69	2.75	-	-	83%	no
GF-08	WC	bath	1.57	2.75	N 0.88	W1 N 1.2x1.54	0%	no
GF-09	BATH	bath	1.63	2.75	W 1.32	-	0%	no
GF-10	COURTYARD	outdoor	11.51	2.75	S 4.37, W 4.13, E	W3A S 0.575x1.175	74%	no

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					1.63			
GF-11	ROOM	bedroom	8.05	2.75	E 2.67	W3A E 0.575x1.175	0%	yes
GF-12	COURTYARD	outdoor	12.30	2.75	E 4.24, S 4.3, W 1.63	-	75%	no
GF-13	COURTYARD	outdoor	12.34	2.75	N 4.26, W 4.25, E 1.62	-	100%	no
GF-14	ROOM	bedroom	7.57	2.75	W 2.68	W3A W 0.575x1.175	0%	yes
GF-15	ROOM	bedroom	8.07	2.75	E 2.67	W3A E 0.575x1.175	0%	yes
GF-16	COURTYARD	outdoor	11.62	2.75	N 4.26, E 4.12, W 1.62	-	100%	no
GF-17	KITCHEN	kitchen	4.22	2.75	N 1.63	W4 N 0.575x1.175	42%	yes
GF-18	BATH	bath	1.59	2.75	E 1.22	-	0%	no
GF-19	WC	bath	1.56	2.75	S 0.89	-	0%	no
GF-20	SHAFT	shaft	2.75	2.75	W 0.54	W1 W 1.2x1.54, W1 W 1.2x1.54	0%	no
GF-21	WC	bath	1.56	2.75	S 0.77	-	0%	no
GF-22	BATH	bath	1.62	2.75	W 1.3	-	0%	no
GF-23	KITCHEN	kitchen	4.24	2.75	N 1.65	W4 N 0.575x1.175	43%	yes
GF-24	ROOM	bedroom	8.12	2.75	S 3.01	-	0%	yes
GF-25	DINNING /LOUNGE	dining	18.44	2.75	N 1.45, S 2.67, W 0.78	W3A S 0.575x1.175, W2 N 0.575x1.175	11%	yes
GF-26	DINNING /LOUNGE	dining	18.40	2.75	N 1.43, E 0.8, S 2.65, W 2.32	W3A S 0.575x1.175, W2 N 0.575x1.175	12%	yes
GF-27	ROOM	bedroom	8.13	2.75	S 2.99	W3A S 0.575x1.175	0%	yes
GF-28	VER.	outdoor	3.69	2.75	S 2.28, W 1.19	W3A W 0.575x1.175	22%	no
GF-29	VER.	outdoor	3.58	2.75	S 3.06	-	23%	no
GF-30	TERRACE	outdoor	6.06	2.75	N 2.39, E 0.66, W 1.9	W3A W 0.575x1.175	100%	no
GF-31	DINNING /LOUNGE	dining	18.13	2.75	N 2.67, S 1.68, W 0.81	W3A N 0.575x1.175	0%	yes
GF-32	DINNING /LOUNGE	dining	18.20	2.75	N 2.68, E 0.79, S 1.68	W3A N 0.575x1.175	23%	yes
GF-33	BATH	bath	1.58	2.75	E 1.18	-	0%	no
GF-34	WC	bath	1.55	2.75	N 0.85	W1 N 1.2x1.54	0%	no
GF-35	SHAFT	shaft	2.67	2.75	-	-	27%	no
GF-36	BATH	bath	1.62	2.75	W 1.32	-	31%	no
GF-37	KITCHEN	kitchen	4.48	2.75	E 2.6, S 1.63	W1 E 1.2x1.54	0%	yes
GF-38	ROOM	bedroom	7.94	2.75	W 2.6	W3A W 0.575x1.175	12%	yes
GF-39	SHAFT	shaft	2.63	2.75	W 0.27	W1 W 1.2x1.54	3%	no
GF-40	TERRACE	outdoor	5.98	2.75	NW 0.09, W 2.29, S 3.01, E 0.81	W3A NW 0.575x1.175	0%	no
GF-41	TERRACE	outdoor	6.38	2.75	E 1.91, S 3.54, W 0.93	W3A E 0.575x1.175	0%	no
FF-01	TERRACE	outdoor	6.06	2.75	N 2.39, E 0.66, W 1.9	W3A W 0.575x1.175	100%	no
FF-02	TERRACE	outdoor	6.54	2.75	N 3.26, E 2.1, W 0.93	W3A E 0.575x1.175	82%	no
FF-03	ROOM	bedroom	8.11	2.75	N 3.0, W 2.6	W1 W 1.2x1.54	5%	yes
FF-04	DINNING /LOUNGE	dining	18.13	2.75	N 2.67, S 1.68, W 0.81	W3A N 0.575x1.175	0%	yes
FF-05	DINNING /LOUNGE	dining	18.20	2.75	N 2.68, E 0.79, S 1.68	W3A N 0.575x1.175	23%	yes
FF-06	ROOM	bedroom	8.13	2.75	N 3.01, E 2.6	W1 E 1.2x1.54	0%	yes
FF-07	KITCHEN	kitchen	4.29	2.75	S 1.63, W 2.6	W1 W 1.2x1.54	0%	yes
FF-08	BATH	bath	1.58	2.75	E 1.18	-	0%	no
FF-09	WC	bath	1.55	2.75	N 0.85	W1 N 1.2x1.54	0%	no
FF-10	SHAFT	shaft	2.67	2.75	-	-	27%	no
FF-11	WC	bath	1.55	2.75	N 0.88	W1 N 1.2x1.54	0%	no
FF-12	BATH	bath	1.62	2.75	W 1.32	-	31%	no
FF-13	KITCHEN	kitchen	4.48	2.75	E 2.6, S 1.63	W1 E 1.2x1.54	0%	yes
FF-14	ROOM	bedroom	7.94	2.75	W 2.6	W3A W 0.575x1.175	12%	yes

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FF-15	ROOM	bedroom	7.94	2.75	E 2.59	W3A E 0.575x1.175	16%	yes
FF-16	ROOM	bedroom	7.98	2.75	W 2.61	W3A W 0.575x1.175	16%	yes
FF-17	ROOM	bedroom	7.96	2.75	E 2.6	W3A E 0.575x1.175	22%	yes
FF-18	KITCHEN	kitchen	4.21	2.75	N 1.64, W 2.58	W4 N 0.575x1.175	63%	yes
FF-19	BATH	bath	1.57	2.75	E 1.22	-	0%	no
FF-20	WC	bath	1.54	2.75	S 0.89	-	0%	no
FF-21	SHAFT	shaft	2.63	2.75	W 0.27	W1 W 1.2x1.54	3%	no
FF-22	WC	bath	1.53	2.75	S 0.77	W1 S 1.2x1.54	44%	no
FF-23	BATH	bath	1.62	2.75	W 1.3	-	0%	no
FF-24	KITCHEN	kitchen	4.21	2.75	N 1.63, E 2.58	W4 N 0.575x1.175, W1 E 1.2x1.54	63%	yes
FF-25	ROOM	bedroom	8.13	2.75	S 2.71, W 2.6	W1 W 1.2x1.54	0%	yes
FF-26	DINNING /LOUNGE	dining	18.01	2.75	S 2.66, W 0.8, N 1.68	W3A S 0.575x1.175, W3A N 0.575x1.175	22%	yes
FF-27	DINNING /LOUNGE	dining	18.14	2.75	N 1.7, E 0.8, S 2.23	W3A S 0.575x1.175, W3A N 0.575x1.175	16%	yes
FF-28	ROOM	bedroom	8.14	2.75	E 2.6, S 2.84	W1 E 1.2x1.54	0%	yes
FF-29	TERRACE	outdoor	5.98	2.75	NW 0.09, W 2.29, S 3.01, E 0.81	W3A NW 0.575x1.175	0%	no
FF-30	TERRACE	outdoor	6.38	2.75	E 1.91, S 3.54, W 0.93	W3A E 0.575x1.175	0%	no
SF-01	TERRACE	outdoor	7.69	2.75	N 2.57, E 0.94, W 2.15	W3A W 0.975x1.175	100%	no
SF-02	TERRACE	outdoor	8.26	2.75	N 3.55, E 2.32, W 0.92	W3A E 0.975x1.175	100%	no
SF-03	ROOM	bedroom	10.38	2.75	N 3.39, W 2.94	-	4%	yes
SF-04	ROOM	bedroom	10.40	2.75	N 3.38, E 1.23	-	4%	yes
SF-05	KITCHEN	kitchen	5.45	2.75	S 1.85, W 2.93	W4 S 0.975x1.175	0%	yes
SF-06	DINNING /LOUNGE	dining	23.07	2.75	N 3.03, S 1.9, W 0.83	W3A S 0.975x1.175	0%	yes
SF-07	BATH	bath	2.04	2.75	E 1.35	-	0%	no
SF-08	SHAFT	shaft	3.37	2.75	-	-	7%	no
SF-09	BATH	bath	2.05	2.75	W 1.32	W1 W 0.575x0.875	0%	no
SF-10	DINNING /LOUNGE	dining	23.03	2.75	N 3.03, E 1.02, S 1.9	W3A N 0.975x1.175, W3A S 0.975x1.175	0%	yes
SF-11	KITCHEN	kitchen	5.74	2.75	S 1.86	-	0%	yes
SF-12	KITCHEN	kitchen	4.90	2.75	N 2.3, E 1.77, SE 1.25	-	100%	yes
SF-13	WC	bath	1.98	2.75	N 1.04	W1 N 0.575x0.875	0%	no
SF-14	WC	bath	1.98	2.75	N 1.0	-	0%	no
SF-15	ROOM	bedroom	5.29	2.75	E 3.16, W 1.05	-	100%	yes
SF-16	ROOM	bedroom	10.18	2.75	W 2.94	W3A W 0.975x1.175	0%	yes
SF-17	ROOM	bedroom	10.13	2.75	E 2.93	W3A E 0.975x1.175	0%	yes
SF-18	OPEN TO SKY	outdoor	13.03	2.75	S 3.89	-	100%	no
SF-19	ROOM	bedroom	10.22	2.75	W 2.95	W3A W 0.975x1.175	0%	yes
SF-20	WC	bath	1.97	2.75	S 1.01	-	0%	no
SF-21	WC	bath	1.96	2.75	S 1.01	-	0%	no
SF-22	ROOM	bedroom	10.18	2.75	E 2.94	W3A E 0.975x1.175	0%	yes
SF-23	KITCHEN	kitchen	5.45	2.75	N 1.84, W 2.93	W4 N 0.975x1.175	4%	yes
SF-24	DINNING /LOUNGE	dining	23.06	2.75	S 2.78, W 0.9, N 1.9	W3A S 0.975x1.175	0%	yes
SF-25	BATH	bath	2.04	2.75	E 1.43	-	0%	no
SF-26	SHAFT	shaft	3.65	2.75	-	-	0%	no
SF-27	BATH	bath	2.08	2.75	W 1.49	-	0%	no
SF-28	DINNING /LOUNGE	dining	23.02	2.75	N 1.9, E 1.06, S 3.03	W3A S 0.975x1.175	0%	yes
SF-29	KITCHEN	kitchen	5.74	2.75	N 1.86, E 2.93	W4 N 0.975x1.175	3%	yes
SF-30	ROOM	bedroom	10.34	2.75	S 2.88, W 2.93	W1 W 0.575x0.875	0%	yes
SF-31	ROOM	bedroom	10.36	2.75	E 2.93, S 2.77	W1 E 0.575x0.875	0%	yes
SF-32	TERRACE	outdoor	7.53	2.75	S 3.39, E 0.91, W 1.67	W3A W 0.975x1.175	100%	no
SF-33	TERRACE	outdoor	8.04	2.75	W 1.19, E 1.77, S 3.84	W3A E 0.975x1.175	100%	no

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SF-34	BALCONY	outdoor	2.33	2.75	N 2.45, E 0.72, W 0.54	-	100%	no
SF-35	BALCONY	outdoor	2.74	2.75	N 1.95, E 1.26, W 0.63	W3A N 0.975x1.175	100%	no
SF-36	BALCONY	outdoor	2.65	2.75	E 0.38, S 2.62, W 0.77	-	100%	no
SF-37	BALCONY	outdoor	2.83	2.75	NE 0.16, W 0.68, E 0.97, S 3.02	W3A NE 0.975x1.175	100%	no
TF-01	TERRACE	outdoor	8.30	2.75	N 3.56, E 0.94, W 2.25	-	100%	no
TF-02	BALCONY	outdoor	2.33	2.75	N 2.45, E 0.72, W 0.54	-	100%	no
TF-03	BALCONY	outdoor	2.74	2.75	N 1.95, E 1.26, W 0.63	W3A N 0.975x1.175	100%	no
TF-04	TERRACE	outdoor	8.72	2.75	N 3.57, E 2.23, W 1.0	W3A E 0.975x1.175	100%	no
TF-05	ROOM	bedroom	10.32	2.75	N 3.43, W 2.95	W3A N 0.975x1.175, W1 W 0.575x0.875	100%	yes
TF-06	ROOM	bedroom	10.36	2.75	N 3.45, E 2.94	W1 E 0.575x0.875	100%	yes
TF-07	KITCHEN	kitchen	5.45	2.75	S 1.85, W 2.93	W4 S 0.975x1.175	100%	yes
TF-08	DINNING /LOUNGE	dining	22.99	2.75	N 1.38, S 1.9, W 0.95	W3A N 0.975x1.175, W3A S 0.975x1.175	100%	yes
TF-09	BATH	bath	2.00	2.75	E 1.33	-	100%	no
TF-10	SHAFT	shaft	3.37	2.75	-	-	100%	no
TF-11	BATH	bath	2.04	2.75	W 1.4	-	100%	no
TF-12	DINNING /LOUNGE	dining	22.93	2.75	N 2.86, E 0.93, S 1.9	W3A S 0.975x1.175	100%	yes
TF-13	KITCHEN	kitchen	5.74	2.75	E 2.93, S 1.86	W4 S 0.975x1.175, W1 E 0.575x0.875	100%	yes
TF-14	WC	bath	1.98	2.75	N 1.04	W1 N 0.575x0.875	100%	no
TF-15	WC	bath	1.98	2.75	N 1.0	W1 N 0.575x0.875	100%	no
TF-16	ROOM	bedroom	10.13	2.75	W 2.94	W3A W 0.975x1.175	100%	yes
TF-17	ROOM	bedroom	10.13	2.75	E 2.93	W3A E 0.975x1.175	100%	yes
TF-18	ROOM	bedroom	10.18	2.75	W 2.95	W3A W 0.975x1.175	100%	yes
TF-19	WC	bath	1.97	2.75	S 1.01	-	100%	no
TF-20	WC	bath	1.96	2.75	S 1.01	-	100%	no
TF-21	ROOM	bedroom	10.18	2.75	E 2.94	W3A E 0.975x1.175	100%	yes
TF-22	KITCHEN	kitchen	5.45	2.75	N 1.85, W 2.93	W4 N 0.975x1.175	100%	yes
TF-23	DINNING /LOUNGE	dining	22.98	2.75	S 2.18, W 0.96, N 1.9	W3A S 0.975x1.175	100%	yes
TF-24	BATH	bath	2.01	2.75	E 1.37	-	100%	no
TF-25	SHAFT	shaft	3.65	2.75	-	-	100%	no
TF-26	BATH	bath	2.08	2.75	W 1.48	-	100%	no
TF-27	DINNING /LOUNGE	dining	22.92	2.75	N 1.9, E 0.96, S 2.5	-	100%	yes
TF-28	KITCHEN	kitchen	5.74	2.75	N 1.86, E 2.93	W4 N 0.975x1.175, W1 E 0.575x0.875	100%	yes
TF-29	ROOM	bedroom	10.29	2.75	S 3.45, W 2.93	W3A S 0.975x1.175, W1 W 0.575x0.875	100%	yes
TF-30	ROOM	bedroom	10.32	2.75	E 2.93, S 3.46	W3A S 0.975x1.175, W1 E 0.575x0.875	100%	yes
TF-31	TERRACE	outdoor	8.35	2.75	E 0.95, S 3.57, W 2.27	-	100%	no
TF-32	BALCONY	outdoor	2.65	2.75	E 0.38, S 2.62, W 0.77	-	100%	no
TF-33	BALCONY	outdoor	2.83	2.75	NE 0.16, W 0.68, E 0.97, S 3.02	W3A NE 0.975x1.175	100%	no
TF-34	TERRACE	outdoor	8.80	2.75	E 2.37, S 3.59, W 1.01	-	100%	no

3. Cooling and heating loads

Method: CLTD/SCL/CLF (ASHRAE Fundamentals 1997 Ch.28). Conduction Q = U·A·CLTDc with CLTDc = (CLTD + LM)·K + (25.5 − Ti) + (Tm − 29.4), Tm = To − DR/2. Glass solar Q = A·SC·SHGF·CLF. People/lights/equipment with CLF = 1.0. Infiltration Qs = Cs·V·ΔT, Ql = Cl·V·ΔW. Loads are computed for solar hours 8-20; each room is sized at its own peak, zones and plant at the coincident (block) peak. Heating: Q = ΣU·A·(Ti − To,w) + infiltration + slab edge, no internal gains.

Code reference: ASHRAE Fundamentals 1997 Ch.28 CLTD/SCL/CLF method, Tables 30-36; ASHRAE Fundamentals 2021 Ch.16 §'Residential air change method'

ID	Room	m²	Walls	Roof	Glass cond.	Glass solar	People S	Lights	Equip S	Infil S	Sensible W	Latent W	Total W	W/m²	Peak	Supply	Heating W
GF-02	ROOM	8.12	55.0	19.0	33.0	21.0	120.0	65.0	41.0	77.0	429	50	479	59.0	15:00	34 L/s	187
GF-03	ROOM	8.13	64.0	21.0	0.0	0.0	120.0	65.0	41.0	72.0	381	50	431	53.1	17:00	30 L/s	164
GF-11	ROOM	8.05	55.0	0.0	23.0	95.0	120.0	64.0	40.0	47.0	445	50	495	61.4	10:00	35 L/s	165
GF-14	ROOM	7.57	70.0	0.0	31.0	104.0	120.0	61.0	38.0	61.0	485	50	535	70.7	18:00	39 L/s	162
GF-15	ROOM	8.07	55.0	0.0	23.0	95.0	120.0	65.0	40.0	47.0	445	50	495	61.3	10:00	35 L/s	165
GF-24	ROOM	8.12	59.0	0.0	0.0	0.0	120.0	65.0	41.0	75.0	360	50	410	50.5	16:00	29 L/s	155
GF-25	DINNING /LOUNGE	18.44	85.0	12.0	66.0	39.0	560.0	184.0	55.0	261.0	1262	360	1622	87.9	15:00	100 L/s	394
GF-26	DINNING /LOUNGE	18.4	147.0	13.0	66.0	38.0	560.0	184.0	55.0	255.0	1317	360	1677	91.2	16:00	105 L/s	473
GF-27	ROOM	8.13	54.0	0.0	33.0	19.0	120.0	65.0	41.0	75.0	407	50	457	56.2	16:00	32 L/s	177
GF-31	DINNING /LOUNGE	18.13	95.0	0.0	33.0	21.0	560.0	181.0	54.0	257.0	1201	360	1561	86.1	15:00	96 L/s	282
GF-32	DINNING /LOUNGE	18.2	102.0	24.0	33.0	21.0	560.0	182.0	55.0	258.0	1233	360	1593	87.5	15:00	98 L/s	295
GF-38	ROOM	7.94	68.0	6.0	31.0	104.0	120.0	64.0	40.0	64.0	496	50	546	68.8	18:00	40 L/s	120
FF-03	ROOM	8.11	122.0	3.0	84.0	286.0	120.0	65.0	41.0	98.0	819	50	869	107.1	18:00	65 L/s	235
FF-04	DINNING /LOUNGE	18.13	95.0	0.0	33.0	21.0	560.0	181.0	54.0	257.0	1201	360	1561	86.1	15:00	96 L/s	282
FF-05	DINNING /LOUNGE	18.2	102.0	24.0	33.0	21.0	560.0	182.0	55.0	258.0	1233	360	1593	87.5	15:00	98 L/s	295
FF-06	ROOM	8.13	91.0	0.0	57.0	293.0	120.0	65.0	41.0	60.0	726	50	776	95.5	9:00	58 L/s	234
FF-14	ROOM	7.94	68.0	6.0	31.0	104.0	120.0	64.0	40.0	64.0	496	50	546	68.8	18:00	40 L/s	120
FF-15	ROOM	7.94	53.0	4.0	23.0	95.0	120.0	64.0	40.0	47.0	445	50	495	62.3	10:00	35 L/s	121
FF-16	ROOM	7.98	68.0	8.0	31.0	104.0	120.0	64.0	40.0	65.0	500	50	550	68.9	18:00	40 L/s	121
FF-17	ROOM	7.96	53.0	6.0	23.0	95.0	120.0	64.0	40.0	47.0	447	50	497	62.4	10:00	36 L/s	122
FF-25	ROOM	8.13	113.0	0.0	84.0	286.0	120.0	65.0	41.0	99.0	807	50	857	105.5	18:00	64 L/s	229
FF-26	DINNING /LOUNGE	18.01	95.0	24.0	66.0	38.0	560.0	180.0	54.0	250.0	1266	360	1626	90.3	16:00	101 L/s	315
FF-27	DINNING /LOUNGE	18.14	89.0	17.0	66.0	39.0	560.0	181.0	54.0	257.0	1262	360	1622	89.4	15:00	100 L/s	306
FF-28	ROOM	8.14	71.0	0.0	57.0	293.0	120.0	65.0	41.0	60.0	707	50	757	92.9	9:00	56 L/s	231
SF-03	ROOM	10.38	158.0	3.0	0.0	0.0	120.0	83.0	52.0	126.0	542	50	592	57.0	18:00	43 L/s	208
SF-04	ROOM	10.4	101.0	2.0	0.0	0.0	120.0	83.0	52.0	147.0	506	50	556	53.4	15:00	40 L/s	179
SF-06	DINNING /LOUNGE	23.07	110.0	0.0	56.0	32.0	700.0	231.0	69.0	320.0	1517	450	1967	85.3	16:00	121 L/s	355
SF-10	DINNING /LOUNGE	23.03	109.0	0.0	111.0	66.0	700.0	230.0	69.0	326.0	1611	450	2061	89.5	15:00	128 L/s	395
SF-15	ROOM	5.29	112.0	32.0	0.0	0.0	120.0	42.0	26.0	73.0	406	50	456	86.2	16:00	32 L/s	138
SF-16	ROOM	10.18	73.0	0.0	52.0	177.0	120.0	81.0	51.0	82.0	637	50	687	67.5	18:00	51 L/s	152
SF-17	ROOM	10.13	52.0	0.0	35.0	182.0	120.0	81.0	51.0	50.0	570	50	620	61.2	9:00	45 L/s	152
SF-19	ROOM	10.22	73.0	0.0	52.0	177.0	120.0	82.0	51.0	83.0	638	50	688	67.3	18:00	51 L/s	153
SF-22	ROOM	10.18	52.0	0.0	35.0	182.0	120.0	81.0	51.0	50.0	571	50	621	61.0	9:00	45 L/s	152

SF-24	DINNING /LOUNGE	23.06	106.0	0.0	56.0	32.0	700.0	231.0	69.0	320.0	1513	450	1963	85.1	16:00	120 L/s	352
SF-28	DINNING /LOUNGE	23.02	116.0	0.0	56.0	31.0	700.0	230.0	69.0	326.0	1528	450	1978	85.9	15:00	122 L/s	358
SF-30	ROOM	10.34	139.0	0.0	23.0	78.0	120.0	83.0	52.0	126.0	619	50	669	64.7	18:00	49 L/s	214
SF-31	ROOM	10.36	129.0	0.0	24.0	22.0	120.0	83.0	52.0	147.0	576	50	626	60.5	15:00	46 L/s	213
TF-05	ROOM	10.32	146.0	64.0	75.0	117.0	120.0	83.0	52.0	125.0	781	50	831	80.6	18:00	62 L/s	291
TF-06	ROOM	10.36	147.0	63.0	24.0	20.0	120.0	83.0	52.0	144.0	653	50	703	67.9	16:00	52 L/s	254
TF-08	DINNING /LOUNGE	22.99	71.0	140.0	111.0	64.0	700.0	230.0	69.0	319.0	1704	450	2154	93.7	16:00	136 L/s	431
TF-12	DINNING /LOUNGE	22.93	114.0	139.0	56.0	32.0	700.0	229.0	69.0	318.0	1657	450	2107	91.9	16:00	132 L/s	418
TF-16	ROOM	10.13	73.0	63.0	52.0	177.0	120.0	81.0	51.0	82.0	699	50	749	73.9	18:00	56 L/s	181
TF-17	ROOM	10.13	57.0	32.0	39.0	161.0	120.0	81.0	51.0	59.0	600	50	650	64.2	10:00	48 L/s	181
TF-18	ROOM	10.18	73.0	63.0	52.0	177.0	120.0	81.0	51.0	82.0	700	50	750	73.7	18:00	56 L/s	182
TF-21	ROOM	10.18	57.0	32.0	39.0	161.0	120.0	81.0	51.0	60.0	602	50	652	64.0	10:00	48 L/s	182
TF-23	DINNING /LOUNGE	22.98	95.0	140.0	56.0	32.0	700.0	230.0	69.0	319.0	1640	450	2090	90.9	16:00	130 L/s	408
TF-27	DINNING /LOUNGE	22.92	114.0	139.0	0.0	0.0	700.0	229.0	69.0	318.0	1569	450	2019	88.1	16:00	125 L/s	375
TF-29	ROOM	10.29	133.0	64.0	78.0	101.0	120.0	82.0	51.0	136.0	765	50	815	79.2	17:00	61 L/s	291
TF-30	ROOM	10.32	138.0	63.0	80.0	52.0	120.0	83.0	52.0	143.0	730	50	780	75.5	16:00	58 L/s	291

Conditioned area 627.5 m²; sum of room peaks 49.8 kW; block peak 48.9 kW at 17:00 (78 W/m²); heating 11.9 kW. Safety factors are applied at equipment selection (section 5).

3.1 Worked example - TF-08 DINNING /LOUNGE at 16:00

Item	Calculation
Wall N	U 0.4 × A 2.99 m² (gross 4.14)
Wall S	U 0.4 × A 4.55 m² (gross 5.7)
Wall W	U 0.4 × A 2.85 m² (gross 2.85)
Window W3A N	cond. U 2.67 × 1.15 m²; solar 1.15 × SC 0.287 × SHGF 130 × CLF
Window W3A S	cond. U 2.67 × 1.15 m²; solar 1.15 × SC 0.287 × SHGF 110 × CLF
Roof	U 0.2 × 22.99 m²
People	10 persons
Infiltration	0.75 ACH × 63.2 m³ = 13.2 L/s
wall	71.0 W
roof	140.0 W
glass_cond	111.0 W
glass_solar	64.0 W
people_s	700.0 W
lights	230.0 W
equip_s	69.0 W
infil_s	319.0 W
people_l	450 W
equip_l	0 W
infil_l	0 W
Room sensible / latent / total	1704 / 450 / 2154 W
Supply airflow	1704 W / (1.143 × (24.0 − 13.0)) = 135.5 L/s

4. Ventilation and exhaust

Ventilation rate procedure: Vbz = Rp·Pz + Ra·Az; Voz = Vbz / Ez; single-zone Vot = Voz; multiple-zone Vot = (D·ΣRp·Pz + ΣRa·Az)/Ev.

Code reference: SBC 501 §403.3.1.1 & Table 403.3.1.1; ASHRAE 62.1-2019 §6.2.2.1 Eq.6-1, Table 6-1; ASHRAE 62.1-2019 §6.2.2.2 Table 6-2 (Ez); ASHRAE 62.1-2019 §6.2.4.3.1 simplified procedure (Ev)

Zone	Room	Category	Pz	Az m²	Rp·Pz	Ra·Az	Vbz L/s	Ez	Ev	Zone Vot L/s	OA method
Z-GF-01	GF-02	dwelling	2	8.12	5.0	2.4	7.4	0.8	1.0	9.3	through-wall fresh air fan
Z-GF-02	GF-03	dwelling	2	8.13	5.0	2.4	7.4	0.8	1.0	9.3	through-wall fresh air fan
Z-GF-03	GF-11	dwelling	2	8.05	5.0	2.4	7.4	0.8	1.0	9.3	through-wall fresh air fan
Z-GF-04	GF-14	dwelling	2	7.57	5.0	2.3	7.3	0.8	1.0	9.1	through-wall fresh air fan
Z-GF-05	GF-15	dwelling	2	8.07	5.0	2.4	7.4	0.8	1.0	9.3	through-wall fresh air fan
Z-GF-06	GF-24	dwelling	2	8.12	5.0	2.4	7.4	0.8	1.0	9.3	through-wall fresh air fan
Z-GF-07	GF-27	dwelling	2	8.13	5.0	2.4	7.4	0.8	1.0	9.3	through-wall fresh air fan
Z-GF-08	GF-38	dwelling	2	7.94	5.0	2.4	7.4	0.8	1.0	9.2	through-wall fresh air fan
Z-GF-09	GF-25	dwelling	8	18.44	20.0	5.5	25.5	0.8	1.0	31.9	ducted to unit return plenum
Z-GF-10	GF-26	dwelling	8	18.4	20.0	5.5	25.5	0.8	1.0	31.9	ducted to unit return plenum
Z-GF-11	GF-31	dwelling	8	18.13	20.0	5.4	25.4	0.8	1.0	31.8	ducted to unit return plenum
Z-GF-12	GF-32	dwelling	8	18.2	20.0	5.5	25.5	0.8	1.0	31.8	ducted to unit return plenum
Z-FF-01	FF-03	dwelling	2	8.11	5.0	2.4	7.4	0.8	1.0	9.3	through-wall fresh air fan
Z-FF-02	FF-06	dwelling	2	8.13	5.0	2.4	7.4	0.8	1.0	9.3	through-wall fresh air fan
Z-FF-03	FF-14	dwelling	2	7.94	5.0	2.4	7.4	0.8	1.0	9.2	through-wall fresh air fan
Z-FF-04	FF-15	dwelling	2	7.94	5.0	2.4	7.4	0.8	1.0	9.2	through-wall fresh air fan
Z-FF-05	FF-16	dwelling	2	7.98	5.0	2.4	7.4	0.8	1.0	9.2	through-wall fresh air fan
Z-FF-06	FF-17	dwelling	2	7.96	5.0	2.4	7.4	0.8	1.0	9.2	through-wall fresh air fan
Z-FF-07	FF-25	dwelling	2	8.13	5.0	2.4	7.4	0.8	1.0	9.3	through-wall fresh air fan
Z-FF-08	FF-28	dwelling	2	8.14	5.0	2.4	7.4	0.8	1.0	9.3	through-wall fresh air fan
Z-FF-09	FF-04	dwelling	8	18.13	20.0	5.4	25.4	0.8	1.0	31.8	ducted to unit return plenum
Z-FF-10	FF-05	dwelling	8	18.2	20.0	5.5	25.5	0.8	1.0	31.8	ducted to unit return plenum
Z-FF-11	FF-26	dwelling	8	18.01	20.0	5.4	25.4	0.8	1.0	31.8	ducted to unit return plenum
Z-FF-12	FF-27	dwelling	8	18.14	20.0	5.4	25.4	0.8	1.0	31.8	ducted to unit return plenum
Z-SF-01	SF-03	dwelling	2	10.38	5.0	3.1	8.1	0.8	1.0	10.1	through-wall fresh air fan
Z-SF-02	SF-04	dwelling	2	10.4	5.0	3.1	8.1	0.8	1.0	10.2	through-wall fresh air fan

Z-SF-03	SF-15	dwelling	2	5.29	5.0	1.6	6.6	0.8	1.0	8.2	through-wall fresh air fan
Z-SF-04	SF-16	dwelling	2	10.18	5.0	3.1	8.1	0.8	1.0	10.1	through-wall fresh air fan
Z-SF-05	SF-17	dwelling	2	10.13	5.0	3.0	8.0	0.8	1.0	10.0	through-wall fresh air fan
Z-SF-06	SF-19	dwelling	2	10.22	5.0	3.1	8.1	0.8	1.0	10.1	through-wall fresh air fan
Z-SF-07	SF-22	dwelling	2	10.18	5.0	3.1	8.1	0.8	1.0	10.1	through-wall fresh air fan
Z-SF-08	SF-30	dwelling	2	10.34	5.0	3.1	8.1	0.8	1.0	10.1	through-wall fresh air fan
Z-SF-09	SF-31	dwelling	2	10.36	5.0	3.1	8.1	0.8	1.0	10.1	through-wall fresh air fan
Z-SF-10	SF-06	dwelling	10	23.07	25.0	6.9	31.9	0.8	1.0	39.9	ducted to unit return plenum
Z-SF-11	SF-10	dwelling	10	23.03	25.0	6.9	31.9	0.8	1.0	39.9	ducted to unit return plenum
Z-SF-12	SF-24	dwelling	10	23.06	25.0	6.9	31.9	0.8	1.0	39.9	ducted to unit return plenum
Z-SF-13	SF-28	dwelling	10	23.02	25.0	6.9	31.9	0.8	1.0	39.9	ducted to unit return plenum
Z-TF-01	TF-05	dwelling	2	10.32	5.0	3.1	8.1	0.8	1.0	10.1	through-wall fresh air fan
Z-TF-02	TF-06	dwelling	2	10.36	5.0	3.1	8.1	0.8	1.0	10.1	through-wall fresh air fan
Z-TF-03	TF-16	dwelling	2	10.13	5.0	3.0	8.0	0.8	1.0	10.0	through-wall fresh air fan
Z-TF-04	TF-17	dwelling	2	10.13	5.0	3.0	8.0	0.8	1.0	10.0	through-wall fresh air fan
Z-TF-05	TF-18	dwelling	2	10.18	5.0	3.1	8.1	0.8	1.0	10.1	through-wall fresh air fan
Z-TF-06	TF-21	dwelling	2	10.18	5.0	3.1	8.1	0.8	1.0	10.1	through-wall fresh air fan
Z-TF-07	TF-29	dwelling	2	10.29	5.0	3.1	8.1	0.8	1.0	10.1	through-wall fresh air fan
Z-TF-08	TF-30	dwelling	2	10.32	5.0	3.1	8.1	0.8	1.0	10.1	through-wall fresh air fan
Z-TF-09	TF-08	dwelling	10	22.99	25.0	6.9	31.9	0.8	1.0	39.9	ducted to unit return plenum
Z-TF-10	TF-12	dwelling	10	22.93	25.0	6.9	31.9	0.8	1.0	39.8	ducted to unit return plenum
Z-TF-11	TF-23	dwelling	10	22.98	25.0	6.9	31.9	0.8	1.0	39.9	ducted to unit return plenum
Z-TF-12	TF-27	dwelling	10	22.92	25.0	6.9	31.9	0.8	1.0	39.8	ducted to unit return plenum

Room	Name	Rule	Exhaust L/s	Mode	Clause
GF-04	KITCHEN	kitchen_residential	50.0	intermittent	SBC 501 Table 403.3.1.1 (kitchens, private dwellings: 50 L/s intermittent or 12 L/s continuous)
GF-05	BATH	toilet_private	25.0	intermittent	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
GF-06	WC	toilet_private	25.0	intermittent	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
GF-08	WC	toilet_private	25.0	intermittent	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)

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GF-09	BATH	toilet_private	25.0	intermittent	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
GF-17	KITCHEN	kitchen_residential	50.0	intermittent	SBC 501 Table 403.3.1.1 (kitchens, private dwellings: 50 L/s intermittent or 12 L/s continuous)
GF-18	BATH	toilet_private	25.0	intermittent	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
GF-19	WC	toilet_private	25.0	intermittent	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
GF-21	WC	toilet_private	25.0	intermittent	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
GF-22	BATH	toilet_private	25.0	intermittent	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
GF-23	KITCHEN	kitchen_residential	50.0	intermittent	SBC 501 Table 403.3.1.1 (kitchens, private dwellings: 50 L/s intermittent or 12 L/s continuous)
GF-33	BATH	toilet_private	25.0	intermittent	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
GF-34	WC	toilet_private	25.0	intermittent	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
GF-36	BATH	toilet_private	25.0	intermittent	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
GF-37	KITCHEN	kitchen_residential	50.0	intermittent	SBC 501 Table 403.3.1.1 (kitchens, private dwellings: 50 L/s intermittent or 12 L/s continuous)
FF-07	KITCHEN	kitchen_residential	50.0	intermittent	SBC 501 Table 403.3.1.1 (kitchens, private dwellings: 50 L/s intermittent or 12 L/s continuous)
FF-08	BATH	toilet_private	25.0	intermittent	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
FF-09	WC	toilet_private	25.0	intermittent	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
FF-11	WC	toilet_private	25.0	intermittent	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
FF-12	BATH	toilet_private	25.0	intermittent	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
FF-13	KITCHEN	kitchen_residential	50.0	intermittent	SBC 501 Table 403.3.1.1 (kitchens, private dwellings: 50 L/s intermittent or 12 L/s continuous)
FF-18	KITCHEN	kitchen_residential	50.0	intermittent	SBC 501 Table 403.3.1.1 (kitchens, private dwellings: 50 L/s intermittent or 12 L/s continuous)
FF-19	BATH	toilet_private	25.0	intermittent	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)

FF-20	WC	toilet_private	25.0	intermittent	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
FF-22	WC	toilet_private	25.0	intermittent	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
FF-23	BATH	toilet_private	25.0	intermittent	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
FF-24	KITCHEN	kitchen_residential	50.0	intermittent	SBC 501 Table 403.3.1.1 (kitchens, private dwellings: 50 L/s intermittent or 12 L/s continuous)
SF-05	KITCHEN	kitchen_residential	50.0	intermittent	SBC 501 Table 403.3.1.1 (kitchens, private dwellings: 50 L/s intermittent or 12 L/s continuous)
SF-07	BATH	toilet_private	25.0	intermittent	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
SF-09	BATH	toilet_private	25.0	intermittent	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
SF-11	KITCHEN	kitchen_residential	50.0	intermittent	SBC 501 Table 403.3.1.1 (kitchens, private dwellings: 50 L/s intermittent or 12 L/s continuous)
SF-12	KITCHEN	kitchen_residential	50.0	intermittent	SBC 501 Table 403.3.1.1 (kitchens, private dwellings: 50 L/s intermittent or 12 L/s continuous)
SF-13	WC	toilet_private	25.0	intermittent	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
SF-14	WC	toilet_private	25.0	intermittent	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
SF-20	WC	toilet_private	25.0	intermittent	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
SF-21	WC	toilet_private	25.0	intermittent	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
SF-23	KITCHEN	kitchen_residential	50.0	intermittent	SBC 501 Table 403.3.1.1 (kitchens, private dwellings: 50 L/s intermittent or 12 L/s continuous)
SF-25	BATH	toilet_private	25.0	intermittent	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
SF-27	BATH	toilet_private	25.0	intermittent	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
SF-29	KITCHEN	kitchen_residential	50.0	intermittent	SBC 501 Table 403.3.1.1 (kitchens, private dwellings: 50 L/s intermittent or 12 L/s continuous)
TF-07	KITCHEN	kitchen_residential	50.0	intermittent	SBC 501 Table 403.3.1.1 (kitchens, private dwellings: 50 L/s intermittent or 12 L/s continuous)
TF-09	BATH	toilet_private	25.0	intermittent	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s

					intermittent or 10 L/s continuous)
TF-11	BATH	toilet_private	25.0	intermittent	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
TF-13	KITCHEN	kitchen_residential	50.0	intermittent	SBC 501 Table 403.3.1.1 (kitchens, private dwellings: 50 L/s intermittent or 12 L/s continuous)
TF-14	WC	toilet_private	25.0	intermittent	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
TF-15	WC	toilet_private	25.0	intermittent	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
TF-19	WC	toilet_private	25.0	intermittent	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
TF-20	WC	toilet_private	25.0	intermittent	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
TF-22	KITCHEN	kitchen_residential	50.0	intermittent	SBC 501 Table 403.3.1.1 (kitchens, private dwellings: 50 L/s intermittent or 12 L/s continuous)
TF-24	BATH	toilet_private	25.0	intermittent	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
TF-26	BATH	toilet_private	25.0	intermittent	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
TF-28	KITCHEN	kitchen_residential	50.0	intermittent	SBC 501 Table 403.3.1.1 (kitchens, private dwellings: 50 L/s intermittent or 12 L/s continuous)

Building outdoor air 891 L/s; exhaust 1,725 L/s (net -834 L/s). ASHRAE 62.2 whole-house cross-check: 213 L/s - design OA exceeds it.

5. Zoning and equipment selection

Required capacity = coincident zone load (rooms + outdoor air) × (1 + SF), SF cooling 10%, heating 10%. Nominal capacities × 0.88 for high-ambient (T3) operation. Selection = smallest catalogue unit meeting total, sensible and airflow.

Code reference: Designer allowance - not a code requirement; SASO 2663 / SBC 601 Table C403.3.2 minimum efficiency

Zone	Rooms	System	Peak h	Rooms kW	OA kW	Block kW	SF	Req. kW	Req. sens kW	Req. air	Selected	Derated kW	Sens kW	Unit air	OA %	Input kW	Heat kW sel/req
Z-GF-01	GF-02	wall_split	15	0.48	0.23	0.71	10%	0.78	0.72	34 L/s	IU-GF-01 WS-09	2.29	1.74	140 L/s	7%	0.76	2.9 / 0.37
Z-GF-02	GF-03	wall_split	15	0.43	0.23	0.66	10%	0.73	0.67	30 L/s	IU-GF-02 WS-09	2.29	1.74	140 L/s	7%	0.76	2.9 / 0.35
Z-GF-03	GF-11	wall_split	15	0.48	0.23	0.71	10%	0.78	0.73	35 L/s	IU-GF-03 WS-09	2.29	1.74	140 L/s	7%	0.76	2.9 / 0.35
Z-GF-04	GF-14	wall_split	17	0.52	0.21	0.73	10%	0.81	0.75	39 L/s	IU-GF-04 WS-09	2.29	1.74	140 L/s	6%	0.76	2.9 / 0.34
Z-GF-05	GF-15	wall_split	15	0.48	0.23	0.71	10%	0.78	0.73	35 L/s	IU-GF-05 WS-09	2.29	1.74	140 L/s	7%	0.76	2.9 / 0.35
Z-GF-06	GF-24	wall_split	15	0.41	0.23	0.64	10%	0.7	0.65	30 L/s	IU-GF-06	2.29	1.74	140 L/s	7%	0.76	2.9 / 0.34

											WS-09						
Z-GF-07	GF-27	wall_split	15	0.46	0.23	0.68	10%	0.75	0.7	32 L/s	IU-GF-07 WS-09	2.29	1.74	140 L/s	7%	0.76	2.9 / 0.36
Z-GF-08	GF-38	wall_split	17	0.54	0.21	0.75	10%	0.82	0.77	40 L/s	IU-GF-08 WS-09	2.29	1.74	140 L/s	7%	0.76	2.9 / 0.3
Z-GF-09	GF-25	ducted_split	15	1.62	0.79	2.41	10%	2.65	2.25	100 L/s	IU-GF-09 DS-18	4.66	3.5	300 L/s	11%	1.66	5.8 / 1.01
Z-GF-10	GF-26	ducted_split	15	1.67	0.79	2.46	10%	2.71	2.31	105 L/s	IU-GF-10 DS-18	4.66	3.5	300 L/s	11%	1.66	5.8 / 1.1
Z-GF-11	GF-31	ducted_split	15	1.56	0.79	2.35	10%	2.58	2.18	96 L/s	IU-GF-11 DS-18	4.66	3.5	300 L/s	11%	1.66	5.8 / 0.89
Z-GF-12	GF-32	ducted_split	15	1.59	0.79	2.38	10%	2.62	2.22	98 L/s	IU-GF-12 DS-18	4.66	3.5	300 L/s	11%	1.66	5.8 / 0.9
Z-FF-01	FF-03	wall_split	18	0.87	0.2	1.07	10%	1.17	1.12	65 L/s	IU-FF-01 WS-09	2.29	1.74	140 L/s	7%	0.76	2.9 / 0.43
Z-FF-02	FF-06	wall_split	10	0.77	0.14	0.91	10%	1.0	0.95	58 L/s	IU-FF-02 WS-09	2.29	1.74	140 L/s	7%	0.76	2.9 / 0.43
Z-FF-03	FF-14	wall_split	17	0.54	0.21	0.75	10%	0.82	0.77	40 L/s	IU-FF-03 WS-09	2.29	1.74	140 L/s	7%	0.76	2.9 / 0.3
Z-FF-04	FF-15	wall_split	15	0.48	0.23	0.71	10%	0.78	0.73	35 L/s	IU-FF-04 WS-09	2.29	1.74	140 L/s	7%	0.76	2.9 / 0.3
Z-FF-05	FF-16	wall_split	17	0.54	0.21	0.75	10%	0.83	0.77	40 L/s	IU-FF-05 WS-09	2.29	1.74	140 L/s	7%	0.76	2.9 / 0.3
Z-FF-06	FF-17	wall_split	15	0.49	0.23	0.71	10%	0.79	0.73	36 L/s	IU-FF-06 WS-09	2.29	1.74	140 L/s	7%	0.76	2.9 / 0.3
Z-FF-07	FF-25	wall_split	18	0.86	0.2	1.05	10%	1.16	1.1	64 L/s	IU-FF-07 WS-09	2.29	1.74	140 L/s	7%	0.76	2.9 / 0.42
Z-FF-08	FF-28	wall_split	15	0.67	0.23	0.9	10%	0.99	0.93	56 L/s	IU-FF-08 WS-09	2.29	1.74	140 L/s	7%	0.76	2.9 / 0.42
Z-FF-09	FF-04	ducted_split	15	1.56	0.79	2.35	10%	2.58	2.18	96 L/s	IU-FF-09 DS-18	4.66	3.5	300 L/s	11%	1.66	5.8 / 0.89
Z-FF-10	FF-05	ducted_split	15	1.59	0.79	2.38	10%	2.62	2.22	98 L/s	IU-FF-10 DS-18	4.66	3.5	300 L/s	11%	1.66	5.8 / 0.9
Z-FF-11	FF-26	ducted_split	15	1.63	0.79	2.41	10%	2.65	2.26	101 L/s	IU-FF-11 DS-18	4.66	3.5	300 L/s	11%	1.66	5.8 / 0.92
Z-FF-12	FF-27	ducted_split	15	1.62	0.79	2.41	10%	2.65	2.25	100 L/s	IU-FF-12 DS-18	4.66	3.5	300 L/s	11%	1.66	5.8 / 0.91
Z-SF-01	SF-03	wall_split	16	0.59	0.24	0.83	10%	0.91	0.86	43 L/s	IU-SF-01 WS-09	2.29	1.74	140 L/s	7%	0.76	2.9 / 0.41
Z-SF-02	SF-04	wall_split	15	0.56	0.25	0.81	10%	0.89	0.83	40 L/s	IU-SF-02 WS-09	2.29	1.74	140 L/s	7%	0.76	2.9 / 0.38
Z-SF-03	SF-15	wall_split	15	0.45	0.2	0.66	10%	0.72	0.67	32 L/s	IU-SF-03 WS-09	2.29	1.74	140 L/s	6%	0.76	2.9 / 0.3
Z-SF-04	SF-16	wall_split	17	0.67	0.23	0.9	10%	0.99	0.94	51 L/s	IU-SF-04 WS-09	2.29	1.74	140 L/s	7%	0.76	2.9 / 0.35
Z-SF-05	SF-17	wall_split	15	0.57	0.25	0.82	10%	0.9	0.85	45 L/s	IU-SF-05 WS-09	2.29	1.74	140 L/s	7%	0.76	2.9 / 0.35
Z-SF-06	SF-19	wall_split	17	0.67	0.23	0.91	10%	1.0	0.94	51 L/s	IU-SF-06 WS-09	2.29	1.74	140 L/s	7%	0.76	2.9 / 0.35
Z-SF-07	SF-22	wall_split	15	0.57	0.25	0.82	10%	0.91	0.85	45 L/s	IU-SF-07 WS-09	2.29	1.74	140 L/s	7%	0.76	2.9 / 0.35
Z-SF-08	SF-30	wall_split	17	0.67	0.23	0.9	10%	0.99	0.93	49 L/s	IU-SF-08 WS-09	2.29	1.74	140 L/s	7%	0.76	2.9 / 0.42
Z-SF-09	SF-31	wall_split	15	0.63	0.25	0.88	10%	0.96	0.91	46 L/s	IU-SF-09 WS-09	2.29	1.74	140 L/s	7%	0.76	2.9 / 0.42
Z-SF-10	SF-06	ducted_split	15	1.97	0.99	2.95	10%	3.25	2.75	121 L/s	IU-SF-10 DS-18	4.66	3.5	300 L/s	13%	1.66	5.8 / 1.11
Z-SF-11	SF-10	ducted_split	15	2.06	0.99	3.05	10%	3.35	2.86	128 L/s	IU-SF-11 DS-18	4.66	3.5	300 L/s	13%	1.66	5.8 / 1.16
Z-SF-12	SF-24	ducted_split	15	1.96	0.99	2.95	10%	3.24	2.75	120 L/s	IU-SF-12 DS-18	4.66	3.5	300 L/s	13%	1.66	5.8 / 1.11
Z-SF-13	SF-28	ducted_split	15	1.98	0.99	2.96	10%	3.26	2.76	122 L/s	IU-SF-13 DS-18	4.66	3.5	300 L/s	13%	1.66	5.8 / 1.12
Z-TF-01	TF-05	wall_split	17	0.82	0.23	1.05	10%	1.16	1.1	62 L/s	IU-TF-01 WS-09	2.29	1.74	140 L/s	7%	0.76	2.9 / 0.5
Z-TF-02	TF-06	wall_split	15	0.7	0.25	0.95	10%	1.05	0.99	52 L/s	IU-TF-02 WS-09	2.29	1.74	140 L/s	7%	0.76	2.9 / 0.46

Z-TF-03	TF-16	wall_split	17	0.73	0.23	0.96	10%	1.06	1.0	56 L/s	IU-TF-03 WS-09	2.29	1.74	140 L/s	7%	0.76	2.9 / 0.38
Z-TF-04	TF-17	wall_split	15	0.63	0.25	0.88	10%	0.97	0.91	48 L/s	IU-TF-04 WS-09	2.29	1.74	140 L/s	7%	0.76	2.9 / 0.38
Z-TF-05	TF-18	wall_split	17	0.73	0.23	0.97	10%	1.06	1.01	56 L/s	IU-TF-05 WS-09	2.29	1.74	140 L/s	7%	0.76	2.9 / 0.38
Z-TF-06	TF-21	wall_split	15	0.63	0.25	0.88	10%	0.97	0.92	48 L/s	IU-TF-06 WS-09	2.29	1.74	140 L/s	7%	0.76	2.9 / 0.38
Z-TF-07	TF-29	wall_split	17	0.81	0.23	1.05	10%	1.15	1.1	61 L/s	IU-TF-07 WS-09	2.29	1.74	140 L/s	7%	0.76	2.9 / 0.5
Z-TF-08	TF-30	wall_split	15	0.78	0.25	1.03	10%	1.13	1.07	58 L/s	IU-TF-08 WS-09	2.29	1.74	140 L/s	7%	0.76	2.9 / 0.5
Z-TF-09	TF-08	ducted_split	15	2.15	0.99	3.14	10%	3.45	2.95	136 L/s	IU-TF-09 DS-18	4.66	3.5	300 L/s	13%	1.66	5.8 / 1.2
Z-TF-10	TF-12	ducted_split	15	2.1	0.98	3.08	10%	3.39	2.9	132 L/s	IU-TF-10 DS-18	4.66	3.5	300 L/s	13%	1.66	5.8 / 1.18
Z-TF-11	TF-23	ducted_split	15	2.08	0.99	3.07	10%	3.37	2.88	130 L/s	IU-TF-11 DS-18	4.66	3.5	300 L/s	13%	1.66	5.8 / 1.17
Z-TF-12	TF-27	ducted_split	15	2.02	0.98	3.0	10%	3.3	2.8	125 L/s	IU-TF-12 DS-18	4.66	3.5	300 L/s	13%	1.66	5.8 / 1.13

Tag	Model	System	Serves	Nominal	Input kW	Location
ODU-01	WS-O-09	wall_split	IU-GF-01	2.6 kW	0.76	GF
ODU-02	WS-O-09	wall_split	IU-GF-02	2.6 kW	0.76	GF
ODU-03	WS-O-09	wall_split	IU-GF-03	2.6 kW	0.76	GF
ODU-04	WS-O-09	wall_split	IU-GF-04	2.6 kW	0.76	GF
ODU-05	WS-O-09	wall_split	IU-GF-05	2.6 kW	0.76	GF
ODU-06	WS-O-09	wall_split	IU-GF-06	2.6 kW	0.76	GF
ODU-07	WS-O-09	wall_split	IU-GF-07	2.6 kW	0.76	GF
ODU-08	WS-O-09	wall_split	IU-GF-08	2.6 kW	0.76	GF
ODU-09	DS-O-18	ducted_split	IU-GF-09	5.3 kW	1.66	RF
ODU-10	DS-O-18	ducted_split	IU-GF-10	5.3 kW	1.66	RF
ODU-11	DS-O-18	ducted_split	IU-GF-11	5.3 kW	1.66	RF
ODU-12	DS-O-18	ducted_split	IU-GF-12	5.3 kW	1.66	RF
ODU-13	WS-O-09	wall_split	IU-FF-01	2.6 kW	0.76	FF
ODU-14	WS-O-09	wall_split	IU-FF-02	2.6 kW	0.76	FF
ODU-15	WS-O-09	wall_split	IU-FF-03	2.6 kW	0.76	FF
ODU-16	WS-O-09	wall_split	IU-FF-04	2.6 kW	0.76	FF
ODU-17	WS-O-09	wall_split	IU-FF-05	2.6 kW	0.76	FF
ODU-18	WS-O-09	wall_split	IU-FF-06	2.6 kW	0.76	FF
ODU-19	WS-O-09	wall_split	IU-FF-07	2.6 kW	0.76	FF
ODU-20	WS-O-09	wall_split	IU-FF-08	2.6 kW	0.76	FF
ODU-21	DS-O-18	ducted_split	IU-FF-09	5.3 kW	1.66	RF
ODU-22	DS-O-18	ducted_split	IU-FF-10	5.3 kW	1.66	RF
ODU-23	DS-O-18	ducted_split	IU-FF-11	5.3 kW	1.66	RF
ODU-24	DS-O-18	ducted_split	IU-FF-12	5.3 kW	1.66	RF
ODU-25	WS-O-09	wall_split	IU-SF-01	2.6 kW	0.76	SF
ODU-26	WS-O-09	wall_split	IU-SF-02	2.6 kW	0.76	SF
ODU-27	WS-O-09	wall_split	IU-SF-03	2.6 kW	0.76	SF
ODU-28	WS-O-09	wall_split	IU-SF-04	2.6 kW	0.76	SF
ODU-29	WS-O-09	wall_split	IU-SF-05	2.6 kW	0.76	SF
ODU-30	WS-O-09	wall_split	IU-SF-06	2.6 kW	0.76	SF
ODU-31	WS-O-09	wall_split	IU-SF-07	2.6 kW	0.76	SF
ODU-32	WS-O-09	wall_split	IU-SF-08	2.6 kW	0.76	SF
ODU-33	WS-O-09	wall_split	IU-SF-09	2.6 kW	0.76	SF
ODU-34	DS-O-18	ducted_split	IU-SF-10	5.3 kW	1.66	RF
ODU-35	DS-O-18	ducted_split	IU-SF-11	5.3 kW	1.66	RF
ODU-36	DS-O-18	ducted_split	IU-SF-12	5.3 kW	1.66	RF

PRELIMINARY DESIGN - FOR COORDINATION ONLY. To be checked and stamped by a qualified HVAC engineer before construction.

ODU-37	DS-O-18	ducted_split	IU-SF-13	5.3 kW	1.66	RF
ODU-38	WS-O-09	wall_split	IU-TF-01	2.6 kW	0.76	TF
ODU-39	WS-O-09	wall_split	IU-TF-02	2.6 kW	0.76	TF
ODU-40	WS-O-09	wall_split	IU-TF-03	2.6 kW	0.76	TF
ODU-41	WS-O-09	wall_split	IU-TF-04	2.6 kW	0.76	TF
ODU-42	WS-O-09	wall_split	IU-TF-05	2.6 kW	0.76	TF
ODU-43	WS-O-09	wall_split	IU-TF-06	2.6 kW	0.76	TF
ODU-44	WS-O-09	wall_split	IU-TF-07	2.6 kW	0.76	TF
ODU-45	WS-O-09	wall_split	IU-TF-08	2.6 kW	0.76	TF
ODU-46	DS-O-18	ducted_split	IU-TF-09	5.3 kW	1.66	RF
ODU-47	DS-O-18	ducted_split	IU-TF-10	5.3 kW	1.66	RF
ODU-48	DS-O-18	ducted_split	IU-TF-11	5.3 kW	1.66	RF
ODU-49	DS-O-18	ducted_split	IU-TF-12	5.3 kW	1.66	RF

6. Duct sizing and air terminals

Equal friction method, target 0.80 Pa/m, velocity limits (m/s) {'main': 5.0, 'branch': 3.5, 'runout': 2.5, 'return': 3.5, 'fresh_air': 4.0, 'exhaust': 4.0}. Darcy-Weisbach with Altshul-Tsal friction factor, roughness 0.09 mm; rectangular sizes from De = 1.30(ab)^0.625/(a+b)^0.25; fittings +50% equivalent length.

Code reference: ASHRAE Fundamentals 2021 Ch.21 (equal friction method, Eq.18-19 Altshul-Tsal); SBC 501 §603.4 metallic ducts; SMACNA 2005 Tables 2-1/2-2 (gauge)

Z-GF-09 - IU-GF-09 (GF)

Segment	Type	Serves	Flow	Size	De mm	Velocity	Friction	Length m	Δp	Gauge
D-GF-001	main	-	300 L/s	300x250	299	4.00 m/s	0.73 Pa/m	0.83	0.9 Pa	26 ga (0.55 mm)
D-GF-002	branch	GF-25	300 L/s	400x250	343	3.00 m/s	0.38 Pa/m	0.22	0.1 Pa	24 ga (0.70 mm)
D-GF-003	runout	GF-25	150 L/s	Ø300 round	300	2.12 m/s	0.21 Pa/m	1.67	0.5 Pa	26 ga (0.55 mm)
D-GF-004	runout	GF-25	150 L/s	Ø300 round	300	2.12 m/s	0.21 Pa/m	1.67	0.5 Pa	26 ga (0.55 mm)
D-GF-005	main	-	300 L/s	400x250	343	3.00 m/s	0.38 Pa/m	1.2	0.7 Pa	24 ga (0.70 mm)
D-GF-006	main	-	32 L/s	Ø125 round	125	2.60 m/s	0.89 Pa/m	2.46	3.3 Pa	26 ga (0.55 mm)

Critical path: required ESP 67 Pa vs available 75 Pa - OK.

Z-GF-10 - IU-GF-10 (GF)

Segment	Type	Serves	Flow	Size	De mm	Velocity	Friction	Length m	Δp	Gauge
D-GF-007	main	-	300 L/s	300x250	299	4.00 m/s	0.73 Pa/m	0.83	0.9 Pa	26 ga (0.55 mm)
D-GF-008	branch	GF-26	300 L/s	400x250	343	3.00 m/s	0.38 Pa/m	0.25	0.1 Pa	24 ga (0.70 mm)
D-GF-009	runout	GF-26	150 L/s	Ø300 round	300	2.12 m/s	0.21 Pa/m	1.67	0.5 Pa	26 ga (0.55 mm)
D-GF-010	runout	GF-26	150 L/s	Ø300 round	300	2.12 m/s	0.21 Pa/m	1.67	0.5 Pa	26 ga (0.55 mm)
D-GF-011	main	-	300 L/s	400x250	343	3.00 m/s	0.38 Pa/m	1.2	0.7 Pa	24 ga (0.70 mm)
D-GF-012	main	-	32 L/s	Ø125 round	125	2.60 m/s	0.89 Pa/m	1.8	2.4 Pa	26 ga (0.55 mm)

Critical path: required ESP 67 Pa vs available 75 Pa - OK.

Z-GF-11 - IU-GF-11 (GF)

Segment	Type	Serves	Flow	Size	De mm	Velocity	Friction	Length m	Δp	Gauge
D-GF-013	main	-	300 L/s	300x250	299	4.00 m/s	0.73 Pa/m	0.82	0.9 Pa	26 ga (0.55 mm)
D-GF-014	branch	GF-31	300 L/s	400x250	343	3.00 m/s	0.38 Pa/m	0.14	0.1 Pa	24 ga (0.70 mm)
D-GF-015	runout	GF-31	150 L/s	Ø300 round	300	2.12 m/s	0.21 Pa/m	1.67	0.5 Pa	26 ga (0.55 mm)
D-GF-016	runout	GF-31	150 L/s	Ø300 round	300	2.12 m/s	0.21 Pa/m	1.67	0.5 Pa	26 ga (0.55 mm)

D-GF-017	main	-	300 L/s	400x250	343	3.00 m/s	0.38 Pa/m	1.2	0.7 Pa	24 ga (0.70 mm)
D-GF-018	main	-	32 L/s	Ø125 round	125	2.59 m/s	0.89 Pa/m	2.19	2.9 Pa	26 ga (0.55 mm)

Critical path: required ESP 67 Pa vs available 75 Pa - OK.

Z-GF-12 - IU-GF-12 (GF)

Segment	Type	Serves	Flow	Size	De mm	Velocity	Friction	Length m	Δp	Gauge
D-GF-019	main	-	300 L/s	300x250	299	4.00 m/s	0.73 Pa/m	0.83	0.9 Pa	26 ga (0.55 mm)
D-GF-020	branch	GF-32	300 L/s	400x250	343	3.00 m/s	0.38 Pa/m	0.16	0.1 Pa	24 ga (0.70 mm)
D-GF-021	runout	GF-32	150 L/s	Ø300 round	300	2.12 m/s	0.21 Pa/m	1.67	0.5 Pa	26 ga (0.55 mm)
D-GF-022	runout	GF-32	150 L/s	Ø300 round	300	2.12 m/s	0.21 Pa/m	1.67	0.5 Pa	26 ga (0.55 mm)
D-GF-023	main	-	300 L/s	400x250	343	3.00 m/s	0.38 Pa/m	1.2	0.7 Pa	24 ga (0.70 mm)
D-GF-024	main	-	32 L/s	Ø125 round	125	2.59 m/s	0.89 Pa/m	2.15	2.9 Pa	26 ga (0.55 mm)

Critical path: required ESP 67 Pa vs available 75 Pa - OK.

Z-FF-09 - IU-FF-09 (FF)

Segment	Type	Serves	Flow	Size	De mm	Velocity	Friction	Length m	Δp	Gauge
D-FF-001	main	-	300 L/s	300x250	299	4.00 m/s	0.73 Pa/m	0.82	0.9 Pa	26 ga (0.55 mm)
D-FF-002	branch	FF-04	300 L/s	400x250	343	3.00 m/s	0.38 Pa/m	0.14	0.1 Pa	24 ga (0.70 mm)
D-FF-003	runout	FF-04	150 L/s	Ø300 round	300	2.12 m/s	0.21 Pa/m	1.67	0.5 Pa	26 ga (0.55 mm)
D-FF-004	runout	FF-04	150 L/s	Ø300 round	300	2.12 m/s	0.21 Pa/m	1.67	0.5 Pa	26 ga (0.55 mm)
D-FF-005	main	-	300 L/s	400x250	343	3.00 m/s	0.38 Pa/m	1.2	0.7 Pa	24 ga (0.70 mm)
D-FF-006	main	-	32 L/s	Ø125 round	125	2.59 m/s	0.89 Pa/m	2.48	3.3 Pa	26 ga (0.55 mm)

Critical path: required ESP 67 Pa vs available 75 Pa - OK.

Z-FF-10 - IU-FF-10 (FF)

Segment	Type	Serves	Flow	Size	De mm	Velocity	Friction	Length m	Δp	Gauge
D-FF-007	main	-	300 L/s	300x250	299	4.00 m/s	0.73 Pa/m	0.83	0.9 Pa	26 ga (0.55 mm)
D-FF-008	branch	FF-05	300 L/s	400x250	343	3.00 m/s	0.38 Pa/m	0.16	0.1 Pa	24 ga (0.70 mm)
D-FF-009	runout	FF-05	150 L/s	Ø300 round	300	2.12 m/s	0.21 Pa/m	1.67	0.5 Pa	26 ga (0.55 mm)
D-FF-010	runout	FF-05	150 L/s	Ø300 round	300	2.12 m/s	0.21 Pa/m	1.67	0.5 Pa	26 ga (0.55 mm)
D-FF-011	main	-	300 L/s	400x250	343	3.00 m/s	0.38 Pa/m	1.2	0.7 Pa	24 ga (0.70 mm)
D-FF-012	main	-	32 L/s	Ø125 round	125	2.59 m/s	0.89 Pa/m	2.48	3.3 Pa	26 ga (0.55 mm)

Critical path: required ESP 67 Pa vs available 75 Pa - OK.

Z-FF-11 - IU-FF-11 (FF)

Segment	Type	Serves	Flow	Size	De mm	Velocity	Friction	Length m	Δp	Gauge
D-FF-013	main	-	300 L/s	300x250	299	4.00 m/s	0.73 Pa/m	0.84	0.9 Pa	26 ga (0.55 mm)
D-FF-014	branch	FF-26	300 L/s	400x250	343	3.00 m/s	0.38 Pa/m	0.15	0.1 Pa	24 ga (0.70 mm)
D-FF-015	runout	FF-26	150 L/s	Ø300 round	300	2.12 m/s	0.21 Pa/m	1.67	0.5 Pa	26 ga (0.55 mm)
D-FF-016	runout	FF-26	150 L/s	Ø300 round	300	2.12 m/s	0.21 Pa/m	1.67	0.5 Pa	26 ga (0.55 mm)
D-FF-017	main	-	300 L/s	400x250	343	3.00 m/s	0.38 Pa/m	1.2	0.7 Pa	24 ga (0.70 mm)
D-FF-018	main	-	32 L/s	Ø125 round	125	2.59 m/s	0.89 Pa/m	2.48	3.3 Pa	26 ga (0.55 mm)

Critical path: required ESP 67 Pa vs available 75 Pa - OK.

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Z-FF-12 - IU-FF-12 (FF)

Segment	Type	Serves	Flow	Size	De mm	Velocity	Friction	Length m	Δp	Gauge
D-FF-019	main	-	300 L/s	300x250	299	4.00 m/s	0.73 Pa/m	0.83	0.9 Pa	26 ga (0.55 mm)
D-FF-020	branch	FF-27	300 L/s	400x250	343	3.00 m/s	0.38 Pa/m	0.16	0.1 Pa	24 ga (0.70 mm)
D-FF-021	runout	FF-27	150 L/s	Ø300 round	300	2.12 m/s	0.21 Pa/m	1.67	0.5 Pa	26 ga (0.55 mm)
D-FF-022	runout	FF-27	150 L/s	Ø300 round	300	2.12 m/s	0.21 Pa/m	1.67	0.5 Pa	26 ga (0.55 mm)
D-FF-023	main	-	300 L/s	400x250	343	3.00 m/s	0.38 Pa/m	1.2	0.7 Pa	24 ga (0.70 mm)
D-FF-024	main	-	32 L/s	Ø125 round	125	2.59 m/s	0.89 Pa/m	2.5	3.3 Pa	26 ga (0.55 mm)

Critical path: required ESP 67 Pa vs available 75 Pa - OK.

Z-SF-10 - IU-SF-10 (SF)

Segment	Type	Serves	Flow	Size	De mm	Velocity	Friction	Length m	Δp	Gauge
D-SF-001	riser	-	300 L/s	300x250	299	4.00 m/s	0.73 Pa/m	1.07	1.2 Pa	26 ga (0.55 mm)
D-SF-002	main	-	300 L/s	300x250	299	4.00 m/s	0.73 Pa/m	0.11	0.1 Pa	26 ga (0.55 mm)
D-SF-003	runout	SF-06	150 L/s	Ø300 round	300	2.12 m/s	0.21 Pa/m	1.89	0.6 Pa	26 ga (0.55 mm)
D-SF-004	runout	SF-06	150 L/s	Ø300 round	300	2.12 m/s	0.21 Pa/m	1.89	0.6 Pa	26 ga (0.55 mm)
D-SF-005	main	-	300 L/s	400x250	343	3.00 m/s	0.38 Pa/m	1.2	0.7 Pa	24 ga (0.70 mm)
D-SF-006	main	-	40 L/s	Ø150 round	150	2.26 m/s	0.55 Pa/m	2.57	2.1 Pa	26 ga (0.55 mm)

Critical path: required ESP 68 Pa vs available 75 Pa - OK.

Z-SF-11 - IU-SF-11 (SF)

Segment	Type	Serves	Flow	Size	De mm	Velocity	Friction	Length m	Δp	Gauge
D-SF-007	riser	-	300 L/s	300x250	299	4.00 m/s	0.73 Pa/m	0.93	1.0 Pa	26 ga (0.55 mm)
D-SF-008	main	-	300 L/s	300x250	299	4.00 m/s	0.73 Pa/m	0.11	0.1 Pa	26 ga (0.55 mm)
D-SF-009	runout	SF-10	150 L/s	Ø300 round	300	2.12 m/s	0.21 Pa/m	1.89	0.6 Pa	26 ga (0.55 mm)
D-SF-010	runout	SF-10	150 L/s	Ø300 round	300	2.12 m/s	0.21 Pa/m	1.89	0.6 Pa	26 ga (0.55 mm)
D-SF-011	main	-	300 L/s	400x250	343	3.00 m/s	0.38 Pa/m	1.2	0.7 Pa	24 ga (0.70 mm)
D-SF-012	main	-	40 L/s	Ø150 round	150	2.26 m/s	0.55 Pa/m	2.44	2.0 Pa	26 ga (0.55 mm)

Critical path: required ESP 67 Pa vs available 75 Pa - OK.

Z-SF-12 - IU-SF-12 (SF)

Segment	Type	Serves	Flow	Size	De mm	Velocity	Friction	Length m	Δp	Gauge
D-SF-013	main	-	300 L/s	300x250	299	4.00 m/s	0.73 Pa/m	0.11	0.1 Pa	26 ga (0.55 mm)
D-SF-014	branch	SF-24	300 L/s	400x250	343	3.00 m/s	0.38 Pa/m	1.07	0.6 Pa	24 ga (0.70 mm)
D-SF-015	runout	SF-24	150 L/s	Ø300 round	300	2.12 m/s	0.21 Pa/m	1.89	0.6 Pa	26 ga (0.55 mm)
D-SF-016	runout	SF-24	150 L/s	Ø300 round	300	2.12 m/s	0.21 Pa/m	1.89	0.6 Pa	26 ga (0.55 mm)
D-SF-017	main	-	300 L/s	400x250	343	3.00 m/s	0.38 Pa/m	1.2	0.7 Pa	24 ga (0.70 mm)
D-SF-018	main	-	40 L/s	Ø150 round	150	2.26 m/s	0.55 Pa/m	2.56	2.1 Pa	26 ga (0.55 mm)

Critical path: required ESP 67 Pa vs available 75 Pa - OK.

Z-SF-13 - IU-SF-13 (SF)

Segment	Type	Serves	Flow	Size	De mm	Velocity	Friction	Length m	Δp	Gauge
D-SF-019	riser	-	300 L/s	300x250	299	4.00 m/s	0.73 Pa/m	0.11	0.1 Pa	26 ga (0.55 mm)
D-SF-020	main	-	300 L/s	300x250	299	4.00 m/s	0.73 Pa/m	0.94	1.0 Pa	26 ga (0.55 mm)
D-SF-021	runout	SF-28	150 L/s	Ø300 round	300	2.12 m/s	0.21 Pa/m	1.89	0.6 Pa	26 ga (0.55 mm)
D-SF-022	runout	SF-28	150 L/s	Ø300 round	300	2.12 m/s	0.21 Pa/m	1.89	0.6 Pa	26 ga (0.55 mm)
D-SF-023	main	-	300 L/s	400x250	343	3.00 m/s	0.38 Pa/m	1.2	0.7 Pa	24 ga (0.70 mm)

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D-SF-024	main	-	40 L/s	Ø150 round	150	2.26 m/s	0.55 Pa/m	2.43	2.0 Pa	26 ga (0.55 mm)

Critical path: required ESP 67 Pa vs available 75 Pa - OK.

Z-TF-09 - IU-TF-09 (TF)

Segment	Type	Serves	Flow	Size	De mm	Velocity	Friction	Length m	Δp	Gauge
D-TF-001	riser	-	300 L/s	300x250	299	4.00 m/s	0.73 Pa/m	1.05	1.2 Pa	26 ga (0.55 mm)
D-TF-002	main	-	300 L/s	300x250	299	4.00 m/s	0.73 Pa/m	0.1	0.1 Pa	26 ga (0.55 mm)
D-TF-003	runout	TF-08	150 L/s	Ø300 round	300	2.12 m/s	0.21 Pa/m	1.91	0.6 Pa	26 ga (0.55 mm)
D-TF-004	runout	TF-08	150 L/s	Ø300 round	300	2.12 m/s	0.21 Pa/m	1.91	0.6 Pa	26 ga (0.55 mm)
D-TF-005	main	-	300 L/s	400x250	343	3.00 m/s	0.38 Pa/m	1.2	0.7 Pa	24 ga (0.70 mm)
D-TF-006	main	-	40 L/s	Ø150 round	150	2.26 m/s	0.55 Pa/m	2.4	2.0 Pa	26 ga (0.55 mm)

Critical path: required ESP 68 Pa vs available 75 Pa - OK.

Z-TF-10 - IU-TF-10 (TF)

Segment	Type	Serves	Flow	Size	De mm	Velocity	Friction	Length m	Δp	Gauge
D-TF-007	riser	-	300 L/s	300x250	299	4.00 m/s	0.73 Pa/m	1.06	1.2 Pa	26 ga (0.55 mm)
D-TF-008	main	-	300 L/s	300x250	299	4.00 m/s	0.73 Pa/m	0.09	0.1 Pa	26 ga (0.55 mm)
D-TF-009	runout	TF-12	150 L/s	Ø300 round	300	2.12 m/s	0.21 Pa/m	1.91	0.6 Pa	26 ga (0.55 mm)
D-TF-010	runout	TF-12	150 L/s	Ø300 round	300	2.12 m/s	0.21 Pa/m	1.91	0.6 Pa	26 ga (0.55 mm)
D-TF-011	main	-	300 L/s	400x250	343	3.00 m/s	0.38 Pa/m	1.2	0.7 Pa	24 ga (0.70 mm)
D-TF-012	main	-	40 L/s	Ø150 round	150	2.25 m/s	0.55 Pa/m	2.4	2.0 Pa	26 ga (0.55 mm)

Critical path: required ESP 68 Pa vs available 75 Pa - OK.

Z-TF-11 - IU-TF-11 (TF)

Segment	Type	Serves	Flow	Size	De mm	Velocity	Friction	Length m	Δp	Gauge
D-TF-013	riser	-	300 L/s	300x250	299	4.00 m/s	0.73 Pa/m	1.07	1.2 Pa	26 ga (0.55 mm)
D-TF-014	main	-	300 L/s	300x250	299	4.00 m/s	0.73 Pa/m	0.1	0.1 Pa	26 ga (0.55 mm)
D-TF-015	runout	TF-23	150 L/s	Ø300 round	300	2.12 m/s	0.21 Pa/m	1.92	0.6 Pa	26 ga (0.55 mm)
D-TF-016	runout	TF-23	150 L/s	Ø300 round	300	2.12 m/s	0.21 Pa/m	1.92	0.6 Pa	26 ga (0.55 mm)
D-TF-017	main	-	300 L/s	400x250	343	3.00 m/s	0.38 Pa/m	1.2	0.7 Pa	24 ga (0.70 mm)
D-TF-018	main	-	40 L/s	Ø150 round	150	2.26 m/s	0.55 Pa/m	2.35	1.9 Pa	26 ga (0.55 mm)

Critical path: required ESP 68 Pa vs available 75 Pa - OK.

Z-TF-12 - IU-TF-12 (TF)

Segment	Type	Serves	Flow	Size	De mm	Velocity	Friction	Length m	Δp	Gauge
D-TF-019	riser	-	300 L/s	300x250	299	4.00 m/s	0.73 Pa/m	1.05	1.2 Pa	26 ga (0.55 mm)
D-TF-020	main	-	300 L/s	300x250	299	4.00 m/s	0.73 Pa/m	0.09	0.1 Pa	26 ga (0.55 mm)
D-TF-021	runout	TF-27	150 L/s	Ø300 round	300	2.12 m/s	0.21 Pa/m	1.92	0.6 Pa	26 ga (0.55 mm)
D-TF-022	runout	TF-27	150 L/s	Ø300 round	300	2.12 m/s	0.21 Pa/m	1.92	0.6 Pa	26 ga (0.55 mm)
D-TF-023	main	-	300 L/s	400x250	343	3.00 m/s	0.38 Pa/m	1.2	0.7 Pa	24 ga (0.70 mm)
D-TF-024	main	-	40 L/s	Ø150 round	150	2.25 m/s	0.55 Pa/m	2.4	2.0 Pa	26 ga (0.55 mm)

Critical path: required ESP 68 Pa vs available 75 Pa - OK.

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6.1 Supply diffuser schedule

Tag	Room	Model	Flow	NC	Throw m
SD-GF-01	GF-25	SD-250	150 L/s	22	2.2
SD-GF-02	GF-25	SD-250	150 L/s	22	2.2
SD-GF-03	GF-26	SD-250	150 L/s	22	2.2
SD-GF-04	GF-26	SD-250	150 L/s	22	2.2
SD-GF-05	GF-31	SD-250	150 L/s	22	2.2
SD-GF-06	GF-31	SD-250	150 L/s	22	2.2
SD-GF-07	GF-32	SD-250	150 L/s	22	2.2
SD-GF-08	GF-32	SD-250	150 L/s	22	2.2
SD-FF-01	FF-04	SD-250	150 L/s	22	2.2
SD-FF-02	FF-04	SD-250	150 L/s	22	2.2
SD-FF-03	FF-05	SD-250	150 L/s	22	2.2
SD-FF-04	FF-05	SD-250	150 L/s	22	2.2
SD-FF-05	FF-26	SD-250	150 L/s	22	2.2
SD-FF-06	FF-26	SD-250	150 L/s	22	2.2
SD-FF-07	FF-27	SD-250	150 L/s	22	2.2
SD-FF-08	FF-27	SD-250	150 L/s	22	2.2
SD-SF-01	SF-06	SD-250	150 L/s	22	2.2
SD-SF-02	SF-06	SD-250	150 L/s	22	2.2
SD-SF-03	SF-10	SD-250	150 L/s	22	2.2
SD-SF-04	SF-10	SD-250	150 L/s	22	2.2
SD-SF-05	SF-24	SD-250	150 L/s	22	2.2
SD-SF-06	SF-24	SD-250	150 L/s	22	2.2
SD-SF-07	SF-28	SD-250	150 L/s	22	2.2
SD-SF-08	SF-28	SD-250	150 L/s	22	2.2
SD-TF-01	TF-08	SD-250	150 L/s	22	2.2
SD-TF-02	TF-08	SD-250	150 L/s	22	2.2
SD-TF-03	TF-12	SD-250	150 L/s	22	2.2
SD-TF-04	TF-12	SD-250	150 L/s	22	2.2
SD-TF-05	TF-23	SD-250	150 L/s	22	2.2
SD-TF-06	TF-23	SD-250	150 L/s	22	2.2
SD-TF-07	TF-27	SD-250	150 L/s	22	2.2
SD-TF-08	TF-27	SD-250	150 L/s	22	2.2

Code reference: ASHRAE Applications 2019 Ch.49 Table 1 (design NC)

6.2 Exhaust systems

Fan	Room	Flow	Model	Duct	Length m	Velocity	ESP req.	Discharge	Clause
EF-GF-01	GF-04	50 L/s	EF-50	Ø150	1.32	2.83 m/s	42 Pa	wall_louver	SBC 501 Table 403.3.1.1 (kitchens, private dwellings: 50 L/s intermittent or 12 L/s continuous)
EF-GF-02	GF-05	25 L/s	EF-25	Ø125	1.37	2.04 m/s	41 Pa	wall_louver	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
EF-GF-03	GF-06	25 L/s	EF-25	Ø125	1.32	2.04 m/s	41 Pa	wall_louver	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)

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EF-GF-04	GF-08	25 L/s	EF-25	Ø125	0.52	2.04 m/s	40 Pa	wall_louver	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
EF-GF-05	GF-09	25 L/s	EF-25	Ø125	2.14	2.04 m/s	42 Pa	wall_louver	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
EF-GF-06	GF-17	50 L/s	EF-50	Ø150	1.28	2.83 m/s	42 Pa	wall_louver	SBC 501 Table 403.3.1.1 (kitchens, private dwellings: 50 L/s intermittent or 12 L/s continuous)
EF-GF-07	GF-18	25 L/s	EF-25	Ø125	0.57	2.04 m/s	40 Pa	wall_louver	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
EF-GF-08	GF-19	25 L/s	EF-25	Ø125	0.52	2.04 m/s	40 Pa	wall_louver	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
EF-GF-09	GF-21	25 L/s	EF-25	Ø125	0.53	2.04 m/s	40 Pa	wall_louver	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
EF-GF-10	GF-22	25 L/s	EF-25	Ø125	0.56	2.04 m/s	40 Pa	wall_louver	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
EF-GF-11	GF-23	50 L/s	EF-50	Ø150	1.27	2.83 m/s	42 Pa	wall_louver	SBC 501 Table 403.3.1.1 (kitchens, private dwellings: 50 L/s intermittent or 12 L/s continuous)
EF-GF-12	GF-33	25 L/s	EF-25	Ø125	1.36	2.04 m/s	41 Pa	wall_louver	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
EF-GF-13	GF-34	25 L/s	EF-25	Ø125	1.33	2.04 m/s	41 Pa	wall_louver	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
EF-GF-14	GF-36	25 L/s	EF-25	Ø125	1.35	2.04 m/s	41 Pa	wall_louver	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s

									intermittent or 10 L/s continuous)
EF-GF-15	GF-37	50 L/s	EF-50	Ø150	0.87	2.83 m/s	41 Pa	wall_louver	SBC 501 Table 403.3.1.1 (kitchens, private dwellings: 50 L/s intermittent or 12 L/s continuous)
EF-FF-01	FF-07	50 L/s	EF-50	Ø150	0.81	2.83 m/s	41 Pa	wall_louver	SBC 501 Table 403.3.1.1 (kitchens, private dwellings: 50 L/s intermittent or 12 L/s continuous)
EF-FF-02	FF-08	25 L/s	EF-25	Ø125	0.56	2.04 m/s	40 Pa	wall_louver	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
EF-FF-03	FF-09	25 L/s	EF-25	Ø125	0.54	2.04 m/s	40 Pa	wall_louver	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
EF-FF-04	FF-11	25 L/s	EF-25	Ø125	0.53	2.04 m/s	40 Pa	wall_louver	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
EF-FF-05	FF-12	25 L/s	EF-25	Ø125	0.55	2.04 m/s	40 Pa	wall_louver	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
EF-FF-06	FF-13	50 L/s	EF-50	Ø150	0.87	2.83 m/s	41 Pa	wall_louver	SBC 501 Table 403.3.1.1 (kitchens, private dwellings: 50 L/s intermittent or 12 L/s continuous)
EF-FF-07	FF-18	50 L/s	EF-50	Ø150	0.81	2.83 m/s	41 Pa	wall_louver	SBC 501 Table 403.3.1.1 (kitchens, private dwellings: 50 L/s intermittent or 12 L/s continuous)
EF-FF-08	FF-19	25 L/s	EF-25	Ø125	0.57	2.04 m/s	40 Pa	wall_louver	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
EF-FF-09	FF-20	25 L/s	EF-25	Ø125	0.52	2.04 m/s	40 Pa	wall_louver	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
EF-FF-10	FF-22	25 L/s	EF-25	Ø125	0.53	2.04 m/s	40 Pa	wall_louver	SBC 501 Table

									403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
EF-FF-11	FF-23	25 L/s	EF-25	Ø125	0.56	2.04 m/s	40 Pa	wall_louver	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
EF-FF-12	FF-24	50 L/s	EF-50	Ø150	0.82	2.83 m/s	41 Pa	wall_louver	SBC 501 Table 403.3.1.1 (kitchens, private dwellings: 50 L/s intermittent or 12 L/s continuous)
EF-SF-01	SF-05	50 L/s	EF-50	Ø150	0.92	2.83 m/s	41 Pa	wall_louver	SBC 501 Table 403.3.1.1 (kitchens, private dwellings: 50 L/s intermittent or 12 L/s continuous)
EF-SF-02	SF-07	25 L/s	EF-25	Ø125	0.64	2.04 m/s	41 Pa	wall_louver	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
EF-SF-03	SF-09	25 L/s	EF-25	Ø125	0.63	2.04 m/s	40 Pa	wall_louver	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
EF-SF-04	SF-11	50 L/s	EF-50	Ø150	1.52	2.83 m/s	42 Pa	wall_louver	SBC 501 Table 403.3.1.1 (kitchens, private dwellings: 50 L/s intermittent or 12 L/s continuous)
EF-SF-05	SF-12	50 L/s	EF-50	Ø150	1.14	2.83 m/s	41 Pa	wall_louver	SBC 501 Table 403.3.1.1 (kitchens, private dwellings: 50 L/s intermittent or 12 L/s continuous)
EF-SF-06	SF-13	25 L/s	EF-25	Ø125	0.61	2.04 m/s	40 Pa	wall_louver	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
EF-SF-07	SF-14	25 L/s	EF-25	Ø125	0.59	2.04 m/s	40 Pa	wall_louver	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
EF-SF-08	SF-20	25 L/s	EF-25	Ø125	0.59	2.04 m/s	40 Pa	wall_louver	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10

									L/s continuous)
EF-SF-09	SF-21	25 L/s	EF-25	Ø125	0.6	2.04 m/s	40 Pa	wall_louver	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
EF-SF-10	SF-23	50 L/s	EF-50	Ø150	0.92	2.83 m/s	41 Pa	wall_louver	SBC 501 Table 403.3.1.1 (kitchens, private dwellings: 50 L/s intermittent or 12 L/s continuous)
EF-SF-11	SF-25	25 L/s	EF-25	Ø125	0.64	2.04 m/s	41 Pa	wall_louver	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
EF-SF-12	SF-27	25 L/s	EF-25	Ø125	0.63	2.04 m/s	40 Pa	wall_louver	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
EF-SF-13	SF-29	50 L/s	EF-50	Ø150	0.98	2.83 m/s	41 Pa	wall_louver	SBC 501 Table 403.3.1.1 (kitchens, private dwellings: 50 L/s intermittent or 12 L/s continuous)
EF-TF-01	TF-07	50 L/s	EF-50	Ø150	0.92	2.83 m/s	41 Pa	wall_louver	SBC 501 Table 403.3.1.1 (kitchens, private dwellings: 50 L/s intermittent or 12 L/s continuous)
EF-TF-02	TF-09	25 L/s	EF-25	Ø125	0.63	2.04 m/s	40 Pa	wall_louver	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
EF-TF-03	TF-11	25 L/s	EF-25	Ø125	0.63	2.04 m/s	40 Pa	wall_louver	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
EF-TF-04	TF-13	50 L/s	EF-50	Ø150	0.98	2.83 m/s	41 Pa	wall_louver	SBC 501 Table 403.3.1.1 (kitchens, private dwellings: 50 L/s intermittent or 12 L/s continuous)
EF-TF-05	TF-14	25 L/s	EF-25	Ø125	0.61	2.04 m/s	40 Pa	wall_louver	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
EF-TF-06	TF-15	25 L/s	EF-25	Ø125	0.59	2.04 m/s	40 Pa	wall_louver	SBC 501 Table 403.3.1.1 (toilet

									rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
EF-TF-07	TF-19	25 L/s	EF-25	Ø125	0.59	2.04 m/s	40 Pa	wall_louver	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
EF-TF-08	TF-20	25 L/s	EF-25	Ø125	0.6	2.04 m/s	40 Pa	wall_louver	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
EF-TF-09	TF-22	50 L/s	EF-50	Ø150	0.92	2.83 m/s	41 Pa	wall_louver	SBC 501 Table 403.3.1.1 (kitchens, private dwellings: 50 L/s intermittent or 12 L/s continuous)
EF-TF-10	TF-24	25 L/s	EF-25	Ø125	0.64	2.04 m/s	41 Pa	wall_louver	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
EF-TF-11	TF-26	25 L/s	EF-25	Ø125	0.63	2.04 m/s	40 Pa	wall_louver	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
EF-TF-12	TF-28	50 L/s	EF-50	Ø150	0.98	2.83 m/s	41 Pa	wall_louver	SBC 501 Table 403.3.1.1 (kitchens, private dwellings: 50 L/s intermittent or 12 L/s continuous)

7. Piping

7.1 Refrigerant piping

Run	Unit → ODU	Capacity	Liquid / gas OD mm	Horiz. floor m	Vertical m	Roof m	Total m	Limit m	Status
RP-IU-GF-01	IU-GF-01 → ODU-01	2.6 kW	6.35 / 9.52 (1/4" / 3/8")	1.47	0.5	0.0	2.2	25	OK
RP-IU-GF-02	IU-GF-02 → ODU-02	2.6 kW	6.35 / 9.52 (1/4" / 3/8")	0.61	0.5	0.0	1.2	25	OK
RP-IU-GF-03	IU-GF-03 → ODU-03	2.6 kW	6.35 / 9.52 (1/4" / 3/8")	1.15	0.5	0.0	1.8	25	OK
RP-IU-GF-04	IU-GF-04 → ODU-04	2.6 kW	6.35 / 9.52 (1/4" / 3/8")	0.62	0.5	0.0	1.2	25	OK
RP-IU-GF-05	IU-GF-05 → ODU-05	2.6 kW	6.35 / 9.52 (1/4" / 3/8")	1.08	0.5	0.0	1.7	25	OK
RP-IU-GF-06	IU-GF-06 → ODU-	2.6 kW	6.35 / 9.52 (1/4" /	0.91	0.5	0.0	1.6	25	OK

	06		3/8")						
RP-IU-GF-07	IU-GF-07 → ODU-07	2.6 kW	6.35 / 9.52 (1/4" / 3/8")	1.49	0.5	0.0	2.2	25	OK
RP-IU-GF-08	IU-GF-08 → ODU-08	2.6 kW	6.35 / 9.52 (1/4" / 3/8")	1.33	0.5	0.0	2.0	25	OK
RP-IU-GF-09	IU-GF-09 → ODU-09	5.3 kW	6.35 / 12.7 (1/4" / 1/2")	4.07	13.5	25.78	47.7	50	OK
RP-IU-GF-10	IU-GF-10 → ODU-10	5.3 kW	6.35 / 12.7 (1/4" / 1/2")	4.08	13.5	27.3	49.4	50	OK
RP-IU-GF-11	IU-GF-11 → ODU-11	5.3 kW	6.35 / 12.7 (1/4" / 1/2")	4.06	13.5	17.85	39.0	50	OK
RP-IU-GF-12	IU-GF-12 → ODU-12	5.3 kW	6.35 / 12.7 (1/4" / 1/2")	4.07	13.5	17.92	39.0	50	OK
RP-IU-FF-01	IU-FF-01 → ODU-13	2.6 kW	6.35 / 9.52 (1/4" / 3/8")	3.64	0.5	0.0	4.6	25	OK
RP-IU-FF-02	IU-FF-02 → ODU-14	2.6 kW	6.35 / 9.52 (1/4" / 3/8")	2.63	0.5	0.0	3.4	25	OK
RP-IU-FF-03	IU-FF-03 → ODU-15	2.6 kW	6.35 / 9.52 (1/4" / 3/8")	1.33	0.5	0.0	2.0	25	OK
RP-IU-FF-04	IU-FF-04 → ODU-16	2.6 kW	6.35 / 9.52 (1/4" / 3/8")	0.61	0.5	0.0	1.2	25	OK
RP-IU-FF-05	IU-FF-05 → ODU-17	2.6 kW	6.35 / 9.52 (1/4" / 3/8")	1.14	0.5	0.0	1.8	25	OK
RP-IU-FF-06	IU-FF-06 → ODU-18	2.6 kW	6.35 / 9.52 (1/4" / 3/8")	1.12	0.5	0.0	1.8	25	OK
RP-IU-FF-07	IU-FF-07 → ODU-19	2.6 kW	6.35 / 9.52 (1/4" / 3/8")	2.61	0.5	0.0	3.4	25	OK
RP-IU-FF-08	IU-FF-08 → ODU-20	2.6 kW	6.35 / 9.52 (1/4" / 3/8")	3.66	0.5	0.0	4.6	25	OK
RP-IU-FF-09	IU-FF-09 → ODU-21	5.3 kW	6.35 / 12.7 (1/4" / 1/2")	5.13	10.5	5.02	22.7	50	OK
RP-IU-FF-10	IU-FF-10 → ODU-22	5.3 kW	6.35 / 12.7 (1/4" / 1/2")	5.16	10.5	10.88	29.2	50	OK
RP-IU-FF-11	IU-FF-11 → ODU-23	5.3 kW	6.35 / 12.7 (1/4" / 1/2")	5.15	10.5	17.73	36.7	50	OK
RP-IU-FF-12	IU-FF-12 → ODU-24	5.3 kW	6.35 / 12.7 (1/4" / 1/2")	5.16	10.5	11.79	30.2	50	OK
RP-IU-SF-01	IU-SF-01 → ODU-25	2.6 kW	6.35 / 9.52 (1/4" / 3/8")	4.09	0.5	0.0	5.0	25	OK
RP-IU-SF-02	IU-SF-02 → ODU-26	2.6 kW	6.35 / 9.52 (1/4" / 3/8")	0.61	0.5	0.0	1.2	25	OK
RP-IU-SF-03	IU-SF-03 → ODU-27	2.6 kW	6.35 / 9.52 (1/4" / 3/8")	2.57	0.5	0.0	3.4	25	OK
RP-IU-SF-04	IU-SF-04 → ODU-28	2.6 kW	6.35 / 9.52 (1/4" / 3/8")	1.19	0.5	0.0	1.9	25	OK
RP-IU-SF-05	IU-SF-05 → ODU-29	2.6 kW	6.35 / 9.52 (1/4" / 3/8")	1.18	0.5	0.0	1.9	25	OK
RP-IU-SF-06	IU-SF-06 → ODU-30	2.6 kW	6.35 / 9.52 (1/4" / 3/8")	1.21	0.5	0.0	1.9	25	OK
RP-IU-SF-07	IU-SF-07 → ODU-31	2.6 kW	6.35 / 9.52 (1/4" / 3/8")	1.17	0.5	0.0	1.8	25	OK
RP-IU-SF-08	IU-SF-08 → ODU-32	2.6 kW	6.35 / 9.52 (1/4" / 3/8")	2.91	0.5	0.0	3.8	25	OK
RP-IU-SF-09	IU-SF-09 → ODU-33	2.6 kW	6.35 / 9.52 (1/4" / 3/8")	4.09	0.5	0.0	5.1	25	OK
RP-IU-SF-10	IU-SF-10 → ODU-34	5.3 kW	6.35 / 12.7 (1/4" / 1/2")	6.05	7.5	18.75	35.5	50	OK
RP-IU-SF-11	IU-SF-11 → ODU-35	5.3 kW	6.35 / 12.7 (1/4" / 1/2")	5.91	7.5	21.25	38.1	50	OK
RP-IU-SF-12	IU-SF-12 → ODU-36	5.3 kW	6.35 / 12.7 (1/4" / 1/2")	6.1	7.5	24.76	42.2	50	OK
RP-IU-SF-13	IU-SF-13 → ODU-	5.3 kW	6.35 / 12.7 (1/4" /	5.98	7.5	21.79	38.8	50	OK

	37		1/2")						
RP-IU-TF-01	IU-TF-01 → ODU-38	2.6 kW	6.35 / 9.52 (1/4" / 3/8")	1.29	0.5	0.0	2.0	25	OK
RP-IU-TF-02	IU-TF-02 → ODU-39	2.6 kW	6.35 / 9.52 (1/4" / 3/8")	1.27	0.5	0.0	2.0	25	OK
RP-IU-TF-03	IU-TF-03 → ODU-40	2.6 kW	6.35 / 9.52 (1/4" / 3/8")	1.19	0.5	0.0	1.9	25	OK
RP-IU-TF-04	IU-TF-04 → ODU-41	2.6 kW	6.35 / 9.52 (1/4" / 3/8")	1.18	0.5	0.0	1.9	25	OK
RP-IU-TF-05	IU-TF-05 → ODU-42	2.6 kW	6.35 / 9.52 (1/4" / 3/8")	1.21	0.5	0.0	1.9	25	OK
RP-IU-TF-06	IU-TF-06 → ODU-43	2.6 kW	6.35 / 9.52 (1/4" / 3/8")	1.17	0.5	0.0	1.8	25	OK
RP-IU-TF-07	IU-TF-07 → ODU-44	2.6 kW	6.35 / 9.52 (1/4" / 3/8")	1.29	0.5	0.0	2.0	25	OK
RP-IU-TF-08	IU-TF-08 → ODU-45	2.6 kW	6.35 / 9.52 (1/4" / 3/8")	1.31	0.5	0.0	2.0	25	OK
RP-IU-TF-09	IU-TF-09 → ODU-46	5.3 kW	6.35 / 12.7 (1/4" / 1/2")	6.09	4.5	10.31	23.0	50	OK
RP-IU-TF-10	IU-TF-10 → ODU-47	5.3 kW	6.35 / 12.7 (1/4" / 1/2")	6.1	4.5	1.4	13.2	50	OK
RP-IU-TF-11	IU-TF-11 → ODU-48	5.3 kW	6.35 / 12.7 (1/4" / 1/2")	6.16	4.5	12.29	25.2	50	OK
RP-IU-TF-12	IU-TF-12 → ODU-49	5.3 kW	6.35 / 12.7 (1/4" / 1/2")	6.15	4.5	13.88	27.0	50	OK

Code reference: SBC 501 Ch.11 §1107 (refrigerant piping); ASHRAE 15; manufacturer piping tables

7.3 Condensate drains

Run	Unit	To	Code min mm	Design mm	Length m	Slope %
CD-IU-GF-01	IU-GF-01	GF-04 KITCHEN (floor drain / waste stack with trap)	20	25	4.51	1.0
CD-IU-GF-02	IU-GF-02	GF-37 KITCHEN (floor drain / waste stack with trap)	20	25	6.23	1.0
CD-IU-GF-03	IU-GF-03	GF-36 BATH (floor drain / waste stack with trap)	20	25	3.42	1.0
CD-IU-GF-04	IU-GF-04	GF-19 WC (floor drain / waste stack with trap)	20	25	4.8	1.0
CD-IU-GF-05	IU-GF-05	GF-21 WC (floor drain / waste stack with trap)	20	25	5.35	1.0
CD-IU-GF-06	IU-GF-06	GF-17 KITCHEN (floor drain / waste stack with trap)	20	25	4.86	1.0
CD-IU-GF-07	IU-GF-07	GF-23 KITCHEN (floor drain / waste stack with trap)	20	25	4.57	1.0
CD-IU-GF-08	IU-GF-08	GF-34 WC (floor drain / waste stack with trap)	20	25	5.66	1.0
CD-IU-GF-09	IU-GF-09	GF-18 BATH (floor drain / waste stack with trap)	20	25	3.76	1.0
CD-IU-GF-10	IU-GF-10	GF-22 BATH (floor drain / waste stack with trap)	20	25	3.76	1.0
CD-IU-GF-11	IU-GF-11	GF-05 BATH (floor drain / waste stack with trap)	20	25	2.55	1.0
CD-IU-GF-12	IU-GF-12	GF-09 BATH (floor drain / waste stack with trap)	20	25	2.56	1.0
CD-IU-FF-01	IU-FF-01	FF-07 KITCHEN (floor drain	20	25	3.22	1.0

		/ waste stack with trap)				
CD-IU-FF-02	IU-FF-02	FF-13 KITCHEN (floor drain / waste stack with trap)	20	25	4.4	1.0
CD-IU-FF-03	IU-FF-03	FF-09 WC (floor drain / waste stack with trap)	20	25	5.66	1.0
CD-IU-FF-04	IU-FF-04	FF-11 WC (floor drain / waste stack with trap)	20	25	4.85	1.0
CD-IU-FF-05	IU-FF-05	FF-20 WC (floor drain / waste stack with trap)	20	25	4.23	1.0
CD-IU-FF-06	IU-FF-06	FF-22 WC (floor drain / waste stack with trap)	20	25	5.39	1.0
CD-IU-FF-07	IU-FF-07	FF-18 KITCHEN (floor drain / waste stack with trap)	20	25	4.41	1.0
CD-IU-FF-08	IU-FF-08	FF-24 KITCHEN (floor drain / waste stack with trap)	20	25	3.27	1.0
CD-IU-FF-09	IU-FF-09	FF-08 BATH (floor drain / waste stack with trap)	20	25	3.73	1.0
CD-IU-FF-10	IU-FF-10	FF-12 BATH (floor drain / waste stack with trap)	20	25	3.76	1.0
CD-IU-FF-11	IU-FF-11	FF-19 BATH (floor drain / waste stack with trap)	20	25	3.75	1.0
CD-IU-FF-12	IU-FF-12	FF-23 BATH (floor drain / waste stack with trap)	20	25	3.77	1.0
CD-IU-SF-01	IU-SF-01	SF-05 KITCHEN (floor drain / waste stack with trap)	20	25	3.67	1.0
CD-IU-SF-02	IU-SF-02	SF-11 KITCHEN (floor drain / waste stack with trap)	20	25	5.73	1.0
CD-IU-SF-03	IU-SF-03	SF-12 KITCHEN (floor drain / waste stack with trap)	20	25	3.43	1.0
CD-IU-SF-04	IU-SF-04	SF-13 WC (floor drain / waste stack with trap)	20	25	6.13	1.0
CD-IU-SF-05	IU-SF-05	SF-14 WC (floor drain / waste stack with trap)	20	25	4.83	1.0
CD-IU-SF-06	IU-SF-06	SF-20 WC (floor drain / waste stack with trap)	20	25	4.81	1.0
CD-IU-SF-07	IU-SF-07	SF-21 WC (floor drain / waste stack with trap)	20	25	6.12	1.0
CD-IU-SF-08	IU-SF-08	SF-23 KITCHEN (floor drain / waste stack with trap)	20	25	4.98	1.0
CD-IU-SF-09	IU-SF-09	SF-29 KITCHEN (floor drain / waste stack with trap)	20	25	3.7	1.0
CD-IU-SF-10	IU-SF-10	SF-07 BATH (floor drain / waste stack with trap)	20	25	4.5	1.0
CD-IU-SF-11	IU-SF-11	SF-09 BATH (floor drain / waste stack with trap)	20	25	4.35	1.0
CD-IU-SF-12	IU-SF-12	SF-25 BATH (floor drain / waste stack with trap)	20	25	4.5	1.0
CD-IU-SF-13	IU-SF-13	SF-27 BATH (floor drain / waste stack with trap)	20	25	4.36	1.0
CD-IU-TF-01	IU-TF-01	TF-07 KITCHEN (floor drain / waste stack with trap)	20	25	5.08	1.0
CD-IU-TF-02	IU-TF-02	TF-13 KITCHEN (floor drain / waste stack with trap)	20	25	6.48	1.0
CD-IU-TF-03	IU-TF-03	TF-14 WC (floor drain / waste stack with trap)	20	25	6.13	1.0
CD-IU-TF-04	IU-TF-04	TF-15 WC (floor drain / waste stack with trap)	20	25	4.84	1.0
CD-IU-TF-05	IU-TF-05	TF-19 WC (floor drain / waste stack with trap)	20	25	4.81	1.0
CD-IU-TF-06	IU-TF-06	TF-20 WC (floor drain / waste stack with trap)	20	25	6.12	1.0
CD-IU-TF-07	IU-TF-07	TF-22 KITCHEN (floor drain	20	25	6.6	1.0

		/ waste stack with trap)				
CD-IU-TF-08	IU-TF-08	TF-28 KITCHEN (floor drain / waste stack with trap)	20	25	4.94	1.0
CD-IU-TF-09	IU-TF-09	TF-09 BATH (floor drain / waste stack with trap)	20	25	4.54	1.0
CD-IU-TF-10	IU-TF-10	TF-11 BATH (floor drain / waste stack with trap)	20	25	4.55	1.0
CD-IU-TF-11	IU-TF-11	TF-24 BATH (floor drain / waste stack with trap)	20	25	4.56	1.0
CD-IU-TF-12	IU-TF-12	TF-26 BATH (floor drain / waste stack with trap)	20	25	4.55	1.0

Code reference: SBC 501 §307.2.1 (disposal), §307.2.2 Table 307.2.2 (drain size), slope >= 1 %

8. Design checks

Check	Status	Detail
Every room served	PASS	49 conditioned rooms served, 52 exhaust rooms; not served by design (stores/shafts): GF-01, GF-07, GF-10, GF-12, GF-13, GF-16, GF-20, GF-28, GF-29, GF-30, GF-35, GF-39, GF-40, GF-41, FF-01, FF-02, FF-10, FF-21, FF-29, FF-30, SF-01, SF-02, SF-08, SF-18, SF-26, SF-32, SF-33, SF-34, SF-35, SF-36, SF-37, TF-01, TF-02, TF-03, TF-04, TF-10, TF-25, TF-31, TF-32, TF-33, TF-34
No duct crossing shaft/stair/column unresolved	PASS	trunk routes chosen clear of shafts, stairs (not served) and columns; beams not on plan - coordinate
Airflows balance	PASS	supply = unit airflow in all ducted zones; building OA 891.3 L/s vs exhaust 1725.0 L/s (net -834 L/s)
All items tagged (unique)	PASS	400 tags
Every drawing item in BOQ	PASS	465 drawn tags all in BOQ (pipe runs 49 measured by size in section G)
Unit external static pressure	PASS	all ducted units have ESP margin
Refrigerant length / height limits	PASS	within generic limits (verify with brand)
Diffuser noise (NC)	PASS	all diffusers at or below room NC target
Fresh-air fraction	PASS	OA fraction within limit for untreated fresh air
Open questions	PASS	none

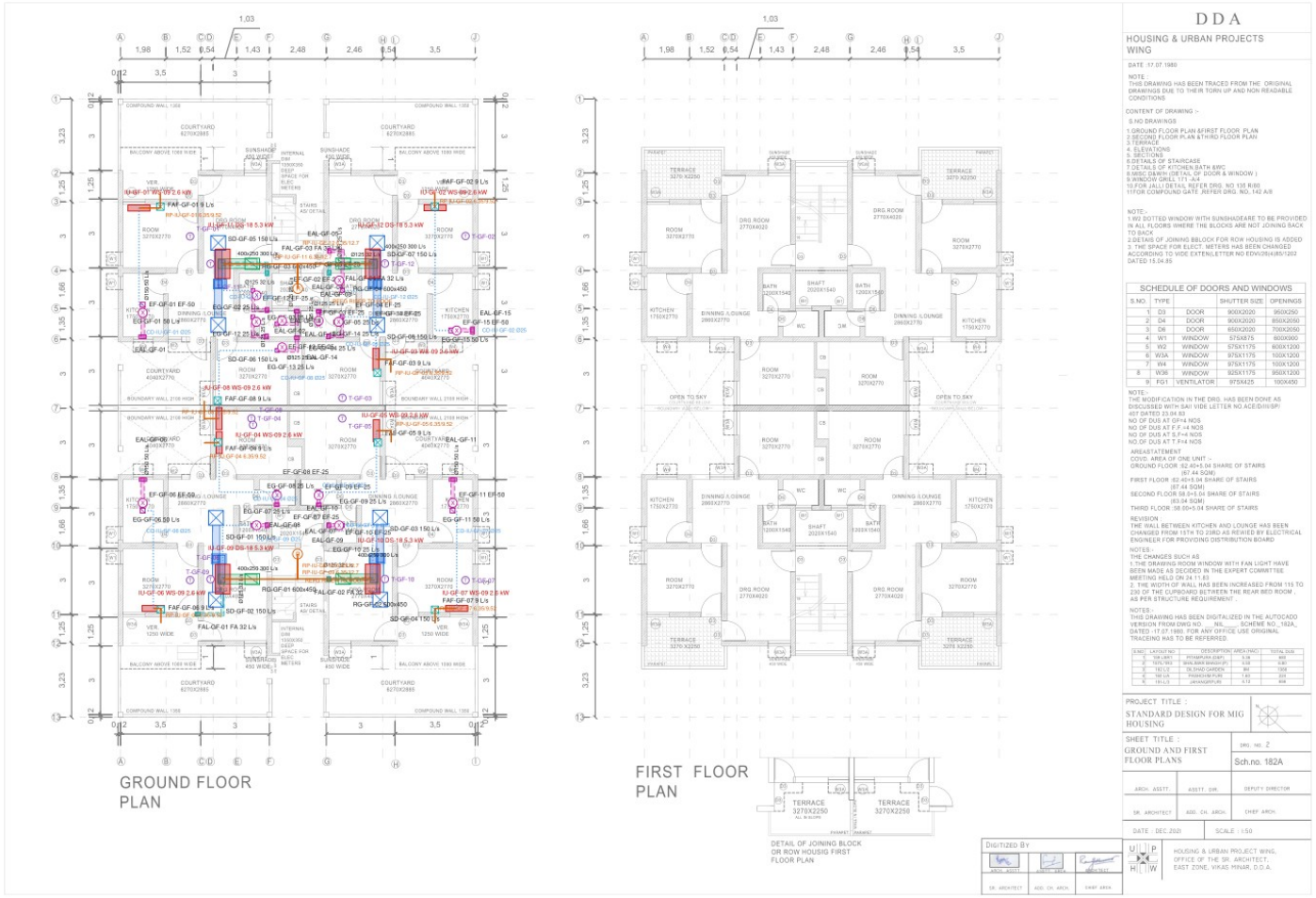
9. Code clause register

Rule	Clause applied
design_conditions	SBC 602 §302 / SBC 601 §302.1 (design conditions); ASHRAE Fundamentals 2021 Ch.14 (0.4 % cooling, 99.6 % heating)
envelope_u	SBC 602 Ch.4 prescriptive envelope table / SBC 601 Ch.5 (max U-factor, SHGC)
load_method	ASHRAE Fundamentals 1997 Ch.28 CLTD/SCL/CLF method, Tables 30-36
infiltration	ASHRAE Fundamentals 2021 Ch.16 §'Residential air change method'
internal_gains	ASHRAE Fundamentals 2021 Ch.18 Table 1 (people), Tables 2-3 (lighting), Tables 5-9 (equipment)
oa_rate	SBC 501 §403.3.1.1 & Table 403.3.1.1; ASHRAE 62.1-2019 §6.2.2.1 Eq.6-1, Table 6-1
oa_zone_eff	ASHRAE 62.1-2019 §6.2.2.2 Table 6-2 (Ez)
oa_system_eff	ASHRAE 62.1-2019 §6.2.4.3.1 simplified procedure (Ev)
exhaust_toilet	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
exhaust_kitchen	SBC 501 Table 403.3.1.1 (kitchens, private dwellings: 50 L/s intermittent or 12 L/s continuous)
exhaust_toilet_public	ASHRAE 62.1-2019 Table 6-4 (toilet public: 25 L/s per WC/urinal)

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duct_sizing	ASHRAE Fundamentals 2021 Ch.21 (equal friction method, Eq.18-19 Altshul-Tsal)
duct_construction	SBC 501 §603.4 metallic ducts; SMACNA 2005 Tables 2-1/2-2 (gauge)
duct_insulation	SBC 601 §C403.11.1 / ASHRAE 90.1-2019 §6.4.4.1.2, Table 6.8.2
pipe_insulation	SBC 601 §C403.11.3 / ASHRAE 90.1-2019 §6.4.4.1.3, Table 6.8.3-1/-2
refrigerant_piping	SBC 501 Ch.11 §1107 (refrigerant piping); ASHRAE 15; manufacturer piping tables
chw_piping	ASHRAE Fundamentals 2021 Ch.22 (water velocity / friction limits)
condensate	SBC 501 §307.2.1 (disposal), §307.2.2 Table 307.2.2 (drain size), slope >= 1 %
equipment_efficiency	SASO 2663 / SBC 601 Table C403.3.2 minimum efficiency
noise	ASHRAE Applications 2019 Ch.49 Table 1 (design NC)
safety_factor	Designer allowance - not a code requirement
controls	SBC 601 §C403.4 / ASHRAE 90.1 §6.4.3 (thermostatic controls, OA damper shut-off)

Appendix - sample sheet (M-101)



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